Informational Notes

Layout Notes

Software Notes

Assembly Notes

Todo Items

Digital

Power

Analog

Comms

Redundant Adaptive VLAN Ethernet Node

Ethernet-to-Fiber Media Converter with Proactive Fault Detection and Redundant Failover

U_Main
Main.SchDoc

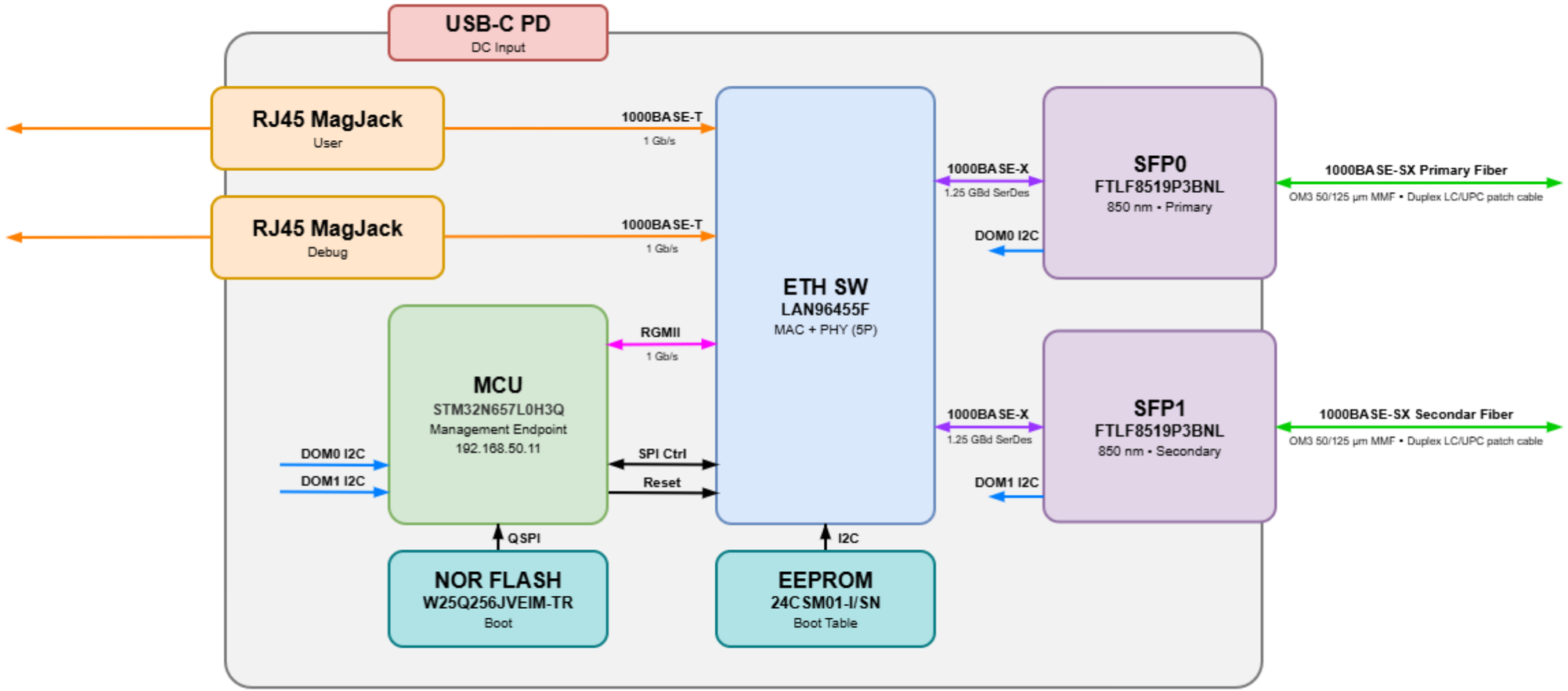
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One-Brd_Block_Diagram.SchDoc

U_System Power Tree
System Power Tree.SchDoc

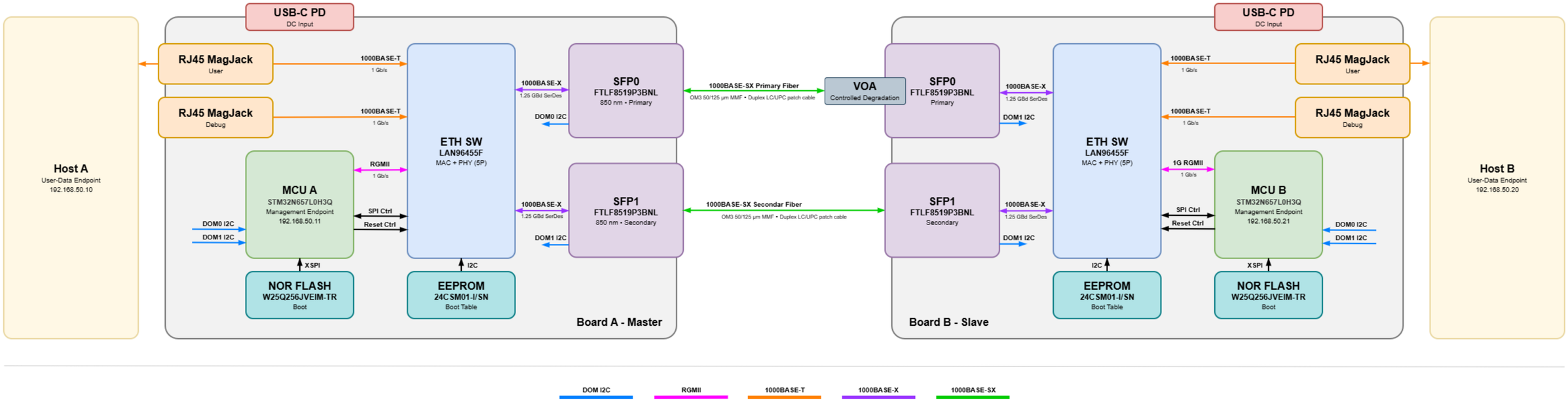
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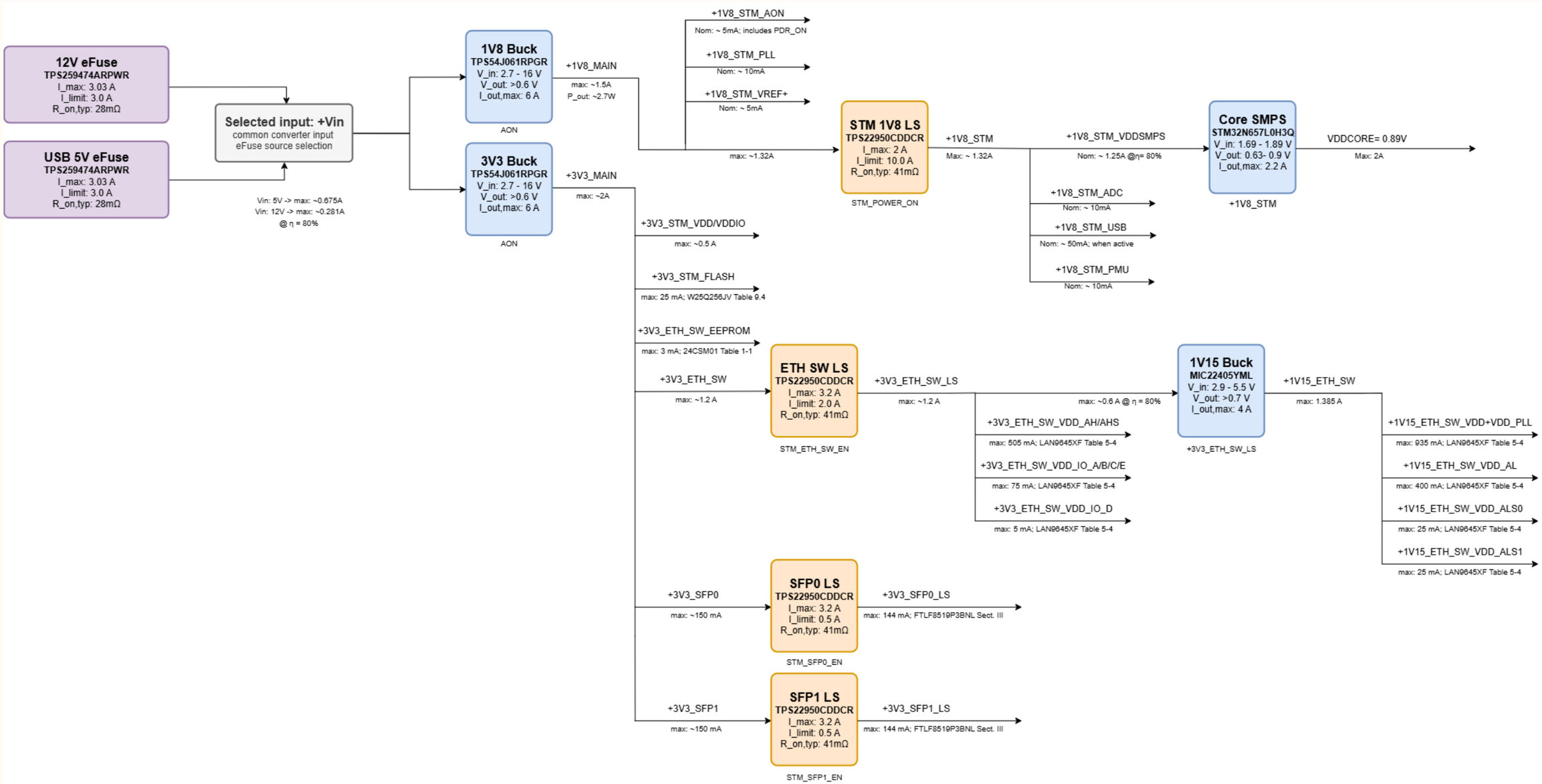
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*
Date: 11/5/2026 Revision: 1.0 Sheet: 1 of: 26		



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Sheet Title:	***One-Brd_Block_Diagram.SchDoc***
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi
Date: 11/5/2026	Revision: 1.0 Sheet: 2 of: 26
Yellow Jacket Space Program	*Georgia Tech* *Atlanta, GA*

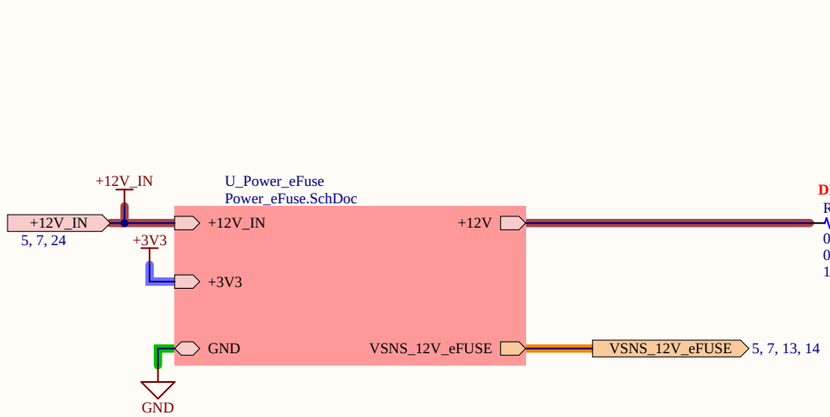


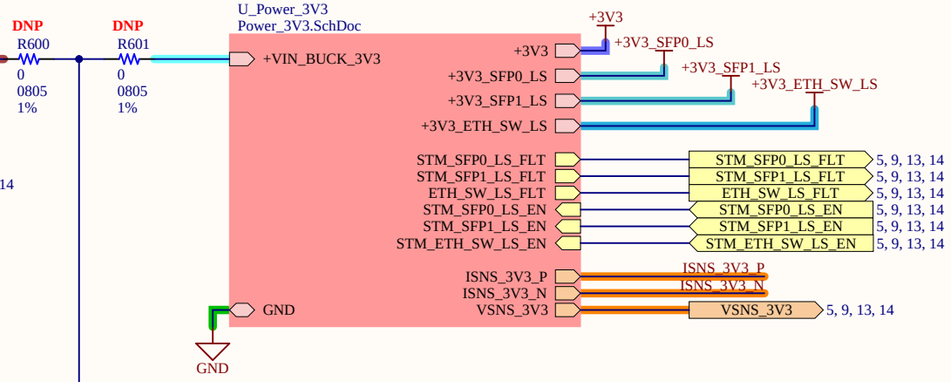
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Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi
Date: 11/5/2026	Revision: 1.0 Sheet: 3 of: 26
Yellow Jacket Space Program *Georgia Tech* *Atlanta, GA*	

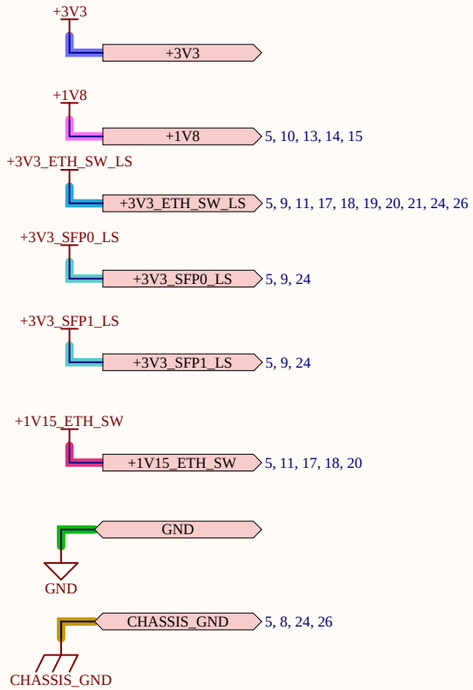


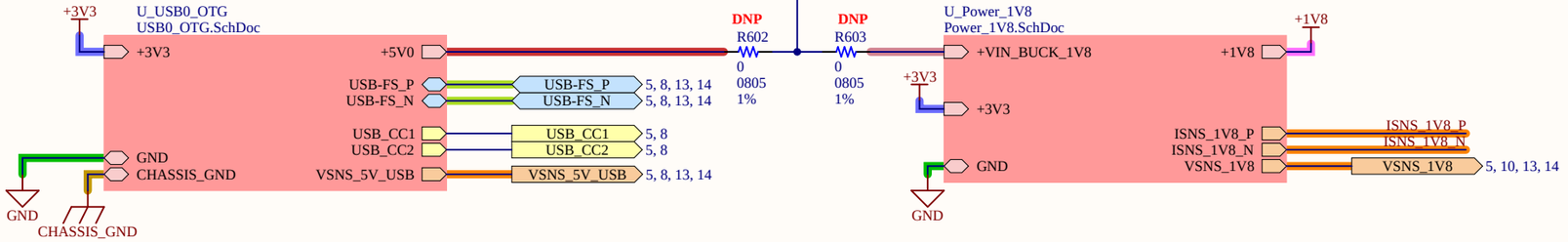
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Sheet Title: ***System Power Tree.SchDoc***		
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*
Date: 11/5/2026 Revision: 1.0	Sheet: 4 of: 26	

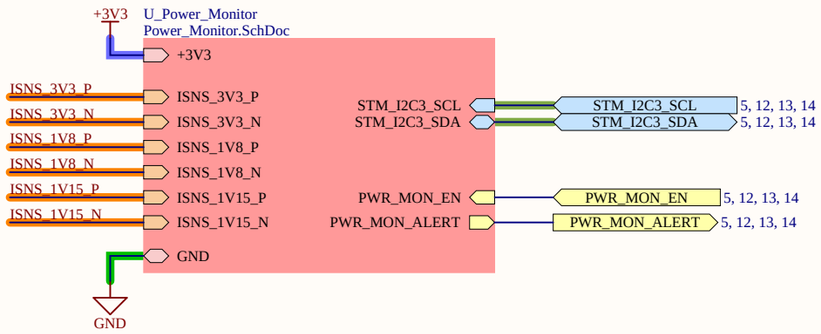
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Power_Top.SchDoc
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+3V3
+1V8
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+3V3_SFP0_LS
+3V3_SFP1_LS
+1V15_ETH_SW
STM_I2C3_SCL
STM_I2C3_SDA
VSNS_3V3
VSNS_1V8
VSNS_1V15
VSNS_5V_USB
VSNS_12V_eFUSE
USB_CC1
USB_CC2
USB-FS_P
USB-FS_N
PWR_MON_ALERT
BUCK_1V15_PG
PWR_MON_EN
STM_1V15_EN
STM_ETH_SW_LS_EN
STM_SFP0_LS_EN
STM_SFP1_LS_EN
ETH_SW_LS_FLT
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STM_SFP1_LS_FLT
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ETH_SW_Top.SchDoc
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Connectors_Top.SchDoc
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STM_SPI4_MISO
STM_SPI4_MOSI
STM_SPI4_CS_N
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STM_ETH_SW_SI_DO
STM_ETH_SW_SI_DI
STM_ETH_SW_SI_NCS
STM_ETH_SW_RESET_N
ETH_SW_EGPIO_29
ETH_SW_EGPIO_46
ETH_SW_IRQ0_SYS_RDYN
VSNS_3V3_ETH_SW_LS
VSNS_3V3_ETH_SW_VDD_AH_SD
VSNS_3V3_ETH_SW_VDD_AH_PHY
VSNS_1V15_ETH_SW
VSNS_1V15_ETH_SW_VDD_PLL
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SFP0_LOS
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SFP0_TX_FAULT_LED
SFP0_MOD_DEF0_LED
SFP1_TX_FAULT
SFP1_MOD_DEF0
SFP1_LOS
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STM_I2C4_SCL
STM_I2C4_SDA
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SFP1_TEMP
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STM_SFP0_MOD_DEF0
STM_SFP0_LOS
STM_SFP0_TX_DIS
STM_SFP0_TX_FAULT_LED
STM_SFP0_MOD_DEF0_LED
STM_SFP1_TX_FAULT
STM_SFP1_MOD_DEF0
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MH501
MH502
MH503
M3_PTH
C500
C501
C502
C503
1210
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10%
10nF
R500
1M
1%
SH500
SH501
CHASSIS_SHIELD_01
CHASSIS_SHIELD_02
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CHASSIS_SHIELD_05
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74737-0010
R501
C504
R502
C505
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Sheet Title: Main.SchDoc
Size: A4
Schematic Designer: Madhawi Alharbi, Ethan Su
PCB Layout Designer: Madhawi Alharbi
Responsible Engineer: Madhawi Alharbi
Date: 11/5/2026
Revision: 1.0
Sheet: 5
of: 26
Yellow Jacket Space Program
Georgia Tech
Atlanta, GA

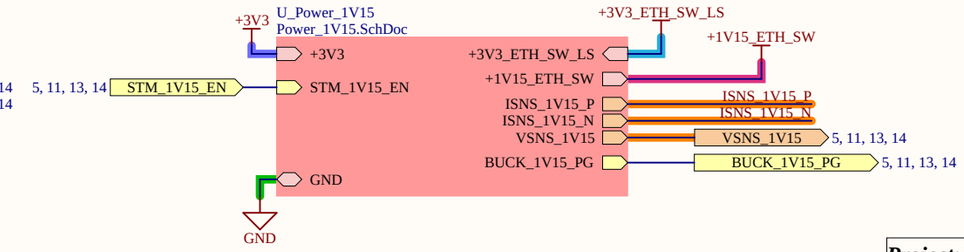




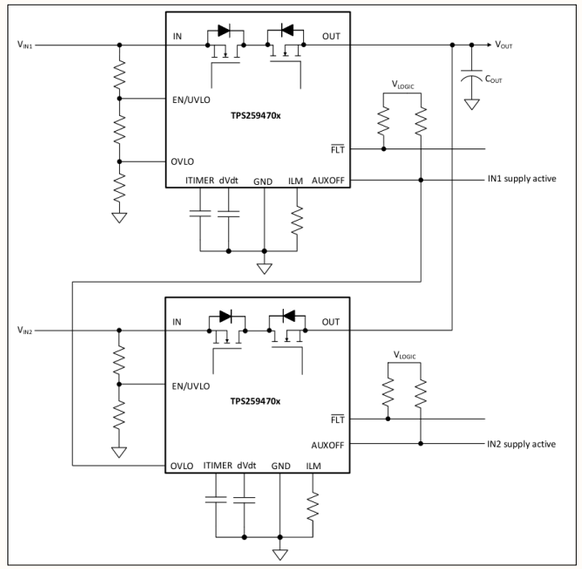








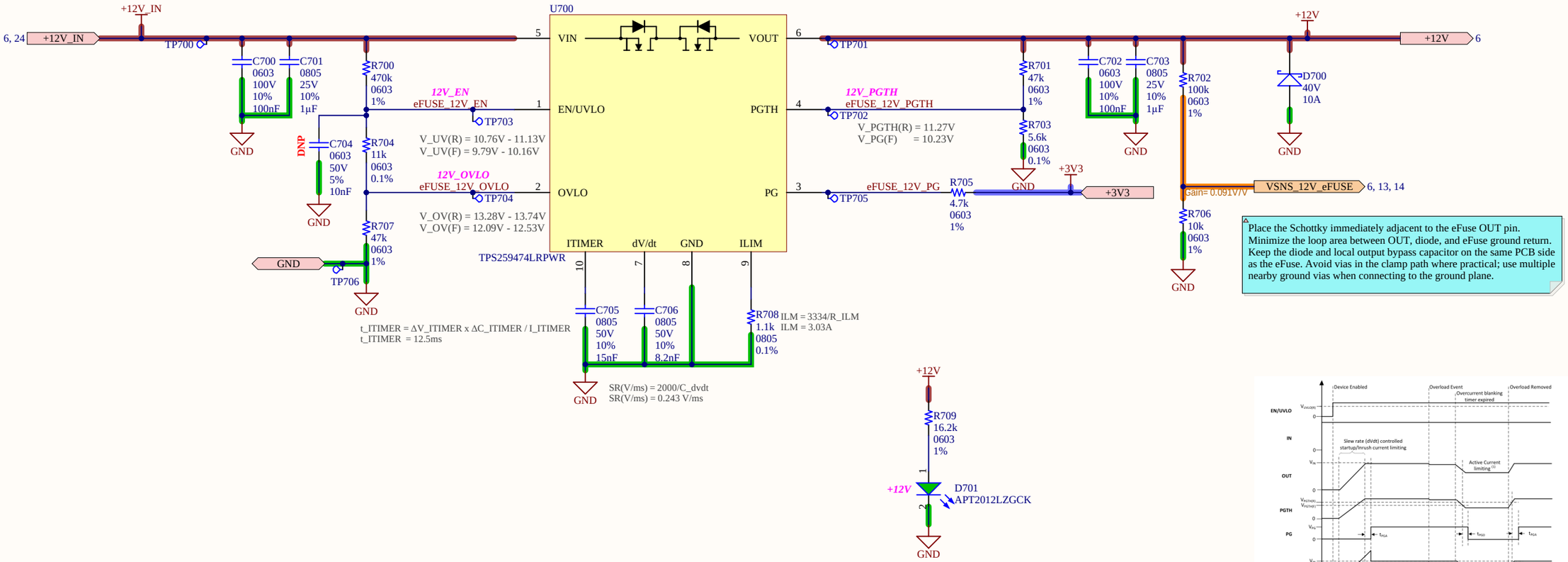
Power MUXing using 2x TPS25947



Project: ***Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb***			
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Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	
Date: 11/5/2026 Revision: 1.0 Sheet: 6 of: 26			

+12V System eFuse

3A Max | Latch-Off



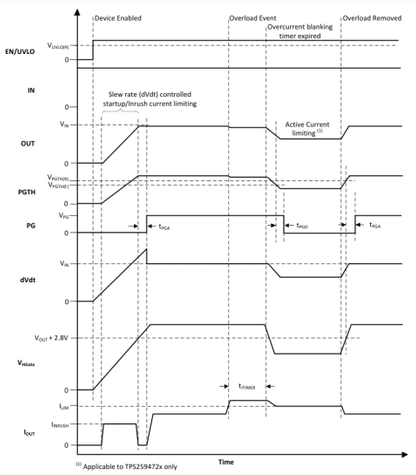


Figure 7-17. TPS259472x, TPS259474x PG Timing Diagram

Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*		
Sheet Title: *Power_eFuse.SchDoc*		
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*
Date: 11/5/2026 Revision: 1.0	Sheet: 7 of: 26	

+5V0 USB eFuse

3A Max | Latch-Off

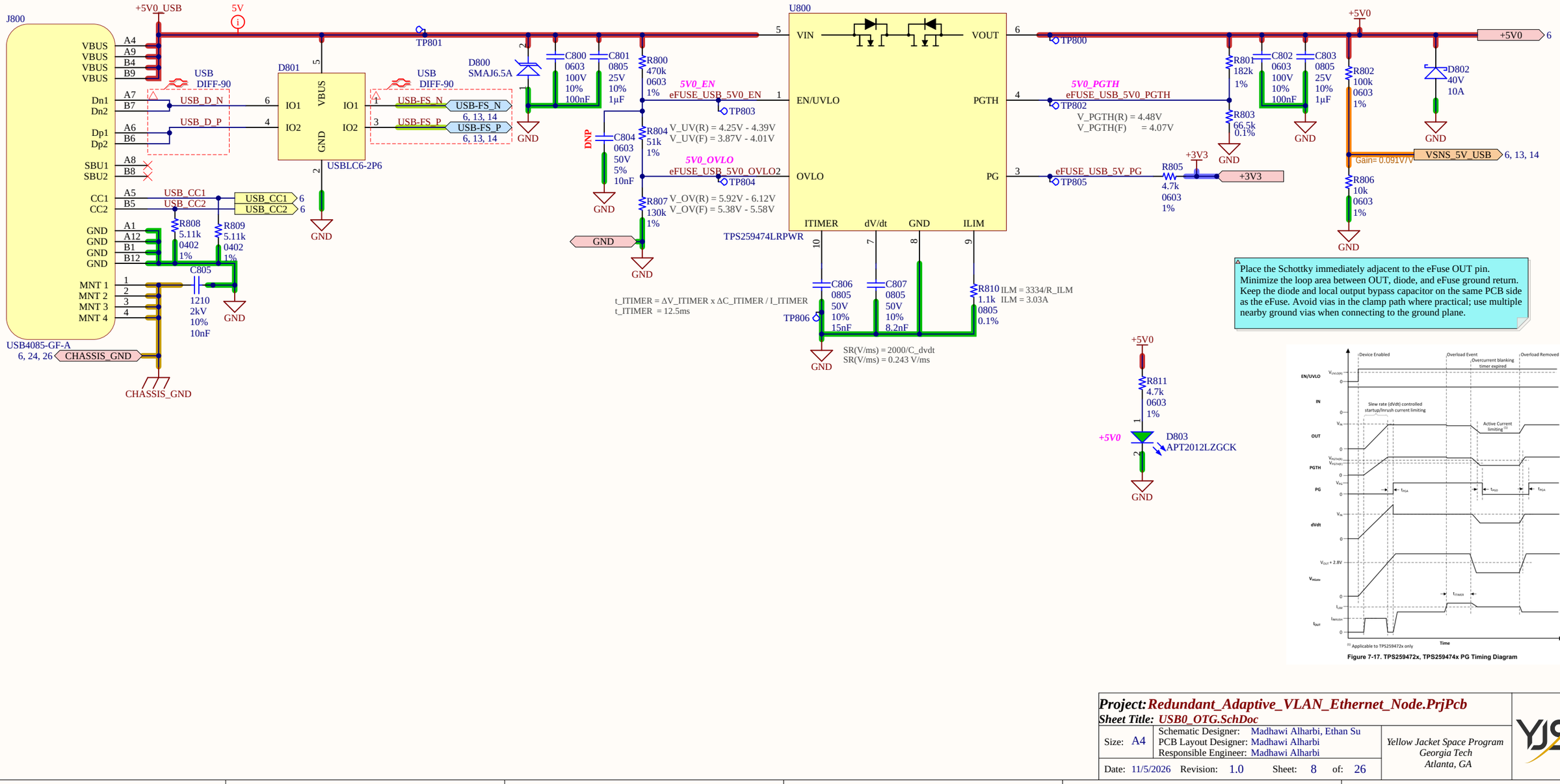


Figure 7-17. TPS259472x, TPS259474x PG Timing Diagram

Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*			
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Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	
Date: 11/5/2026	Revision: 1.0	Sheet: 8 of: 26	

+3V3 Buck Regulator

+12V/+5V0 (16V Max) -> +3V3 / 3A (6A Max)

than the VCC_{UVLO} rising threshold. A practical example is: VIN applied first, then the external bias applied, and then EN signal goes high. When the EN pin voltage rises above the enable threshold voltage, the device enters the start-up sequence as above. A good power-down sequence is the reverse, where either the VIN_{UVLO} falling threshold or EN falling threshold is satisfied before the VCC_{UVLO} falling threshold.

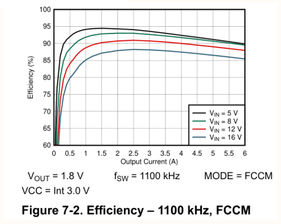


Figure 7-2. Efficiency – 1100 kHz, FCCM

Use separate traces to connect SW node to the bootstrap capacitor and RC snubber instead of combining them into one connection. Avoid using vias for the RC snubber routing and use very wide traces.

To be most effective, the RC snubber must be connected between a large SW copper shape and large PGND copper shape on the same side of the PCB as the TPS54J061.

normal due to parasitic inductance in PCB layout. Including a 0-Ω BOOT resistor is recommended with external VCC as the SW node overshoot is increased. The recommended BOOT resistor value to decrease the SW pin overshoot is 4.7 Ω.

Keep the switch node connections from pins 2 and 11 to the inductor as short and wide as possible.

Inductor substitutions require review of inductance, saturation current, RMS current rating, DCR, and footprint.

Populate all fitted 0 Ω configuration links. Do not replace a DNP resistor with a solder bridge.

All sensitive analog traces FB, PGOOD, TRIP, MODE, and SS/REFIN must be placed away from high-voltage switching nodes such as SW and BOOT to avoid coupling.

Avoid connecting AGND to the PCB ground plane (PGND) in a high current path where significant IR and L*dI/dt drops can occur.

```
3V3 BUCK – NOMINAL CALCULATIONS
VOUT = 0.6*[1+(43.2k+1.8K)/10k] = 3.3V.
UVP: 80% typ = 2.640V; threshold range 2.541–2.739V.
OVP: 116% typ = 3.828V; range 3.729–3.927V.
OOB: early non-latching correction at 105% typ = 3.465V.
MODE low: 1.1 MHz FCCM.
At VIN=5 V, L=2.2uH: inductor ripple ≈0.9886 A p-p.
RTRIP=9.09k: valley limit ≈3.3003 A;
average current-limit onset ≈3.5321 A.
CSS=1nF: soft-start ramp ≈1.5 ms,
excluding startup delays.

Ref: TI SLVSFQ0B, §§6.3.3–6.3.4 p.12,
Table 6-1 p.13, §6.3.7 p.15.
```

+3V3 SFP0/1 Load Switch

500mA Limit

SFP 3V3 LOAD SWITCH
RILIM=2.21kohm: typical current limit ≈504mA.
Estimated Max operating load ≈144mA,

ON: MCU-controlled, default LOW.
FLT: active-low open-drain FAULT

Ref: TI SLVSFJ2B, Table6-1 p.4,
§10.2.2 Eq4 p.16, §§10.3–10.4.1 p.17.

+3V3 ETH SW Load Switch

2A Limit

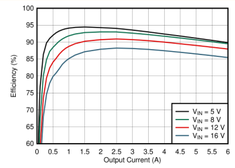
```
ETHERNET 3V3 LOAD SWITCH
RILIM=604ohm: typical current limit ≈2.03A.
Estimated Max operating load ≈1.19A,

ON: MCU-controlled, default LOW.
FLT: active-low open-drain FAULT

Ref: TI SLVSFJ2B, Table6-1 p.4,
§10.2.2 Eq4 p.16, §§10.3–10.4.1 p.17.
```

Project: ***Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb***			
Sheet Title: ***Power_3V3.SchDoc***			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	
Date: 11/5/2026 Revision: 1.0	Sheet: 9 of: 26		

than the VCC_{UVLO} rising threshold. A practical example is: VIN applied first, then the external bias applied, and then EN signal goes high. When the EN pin voltage rises above the enable threshold voltage, the device enters the start-up sequence as above. A good power-down sequence is the reverse, where either the VIN_{UVLO} falling threshold or EN falling threshold is satisfied before the VCC_{UVLO} falling threshold.



V_{OUT} = 1.8 V f_{SW} = 1100 kHz MODE = FCCM
VCC = Int 3.0 V

Figure 7-2. Efficiency – 1100 kHz, FCCM

+1V8 Buck Regulator

+12V/+5V0 (16V Max) -> +1V8 / 2A (6A Max)

Use separate traces to connect SW node to the bootstrap capacitor and RC snubber instead of combining them into one connection. Avoid using vias for the RC snubber routing and use very wide traces.

To be most effective, the RC snubber must be connected between a large SW copper shape and large PGND copper shape on the same side of the PCB as the TPS54J061.

Inductor substitutions require review of inductance, saturation current, RMS current rating, DCR, and footprint.

Keep the switch node connections from pins 2 and 11 to the inductor as short and wide as possible.

Populate all fitted 0 Ω configuration links. Do not replace a DNP resistor with a solder bridge.

All sensitive analog traces FB, PGOOD, TRIP, MODE, and SS/REFIN must be placed away from high-voltage switching nodes such as SW and BOOT to avoid coupling.

Avoid connecting AGND to the PCB ground plane (PGND) in a high current path where significant IR and L*dI/dt drops can occur.

```
1V8 BUCK — NOMINAL CALCULATIONS
VOUT = 0.6*[1+(20k)/10k] = 1.8V.
UVP: 80% typ (~0.927V); threshold range 77–83%.
OVP: 116% typ (~1.344V); threshold range 113–119%.
OOB: early non-latching correction at 105% typ (~1.217V).
MODE low: 1.1 MHz FCCM.
At VIN=5 V, L=2.2uH: inductor ripple ≈0.6322 A p-p.
RTRIP=14k: valley limit ≈2.1429 A;
average current-limit onset ≈2.3809 A.
CSS=1nF: soft-start ramp ≈1.5 ms,
excluding startup delays.

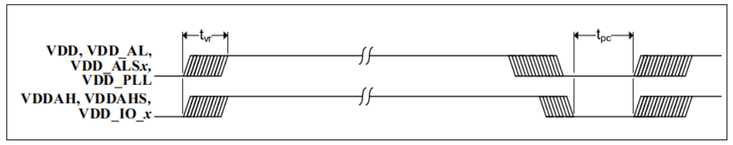
Ref: TI SLVSFQ0B, §§6.3.3–6.3.4 p.12,
Table 6-1 p.13, §6.3.7 p.15.
```

Project:	Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb		
Sheet Title:	Power_1V8.SchDoc		
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	Yellow Jacket Space Program Georgia Tech Atlanta, GA	
Date: 11/5/2026	Revision: 1.0	Sheet: 10 of: 26	

5.3 Power Sequence Timing

This diagram illustrates the device power sequencing requirements.

FIGURE 5-1: POWER SEQUENCE TIMING



All supplies must be powered up within 100 ms (t_{vr}) of each other.

It is recommended to have transceiver (**VDD_AH**) and digital I/O (**VDDIO_IO_*x***) voltages power up before the 1.15V core (**VDD, VDD_AL, VDD_ALSx, VDD_PLL**) voltage. If the 1.15V core must power up first, the maximum lead time for the 1.15V core voltage with respect to the transceiver and digital I/O voltages must be 200 µs.

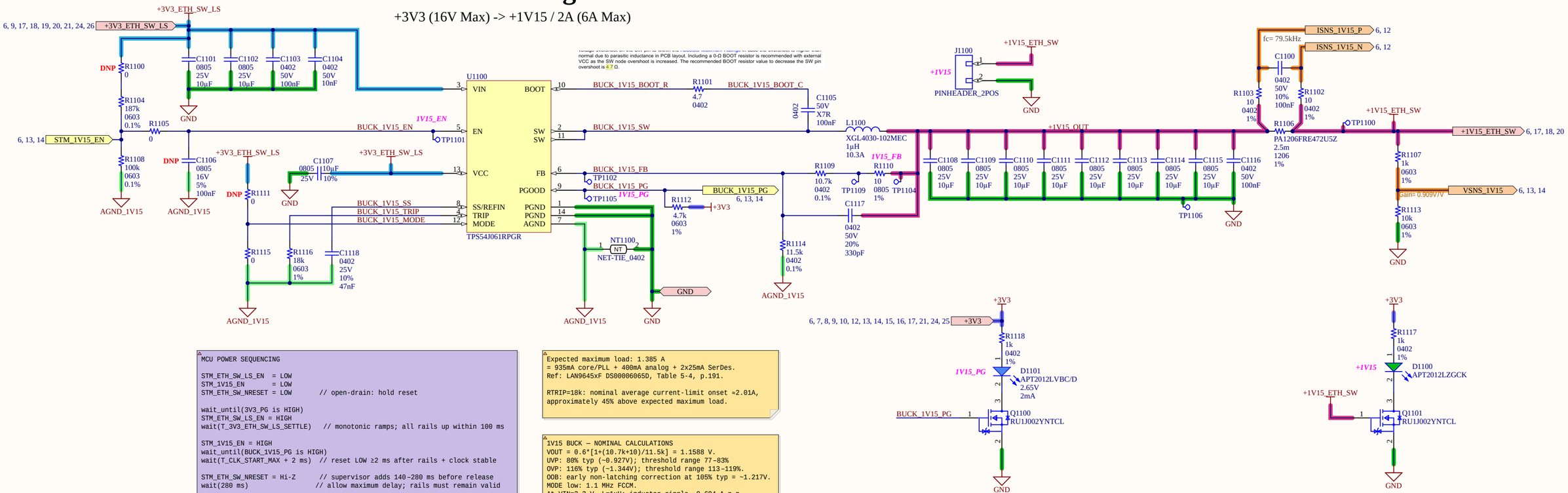
The power-up waveforms must be monotonic for all supply voltages to the device.

TABLE 5-1: POWER SEQUENCING TIMING VALUES

Symbol	Description	Min	Typ	Max	Unit
t_{vr}	Supply voltages rise time (must be monotonic)	0.1	—	100	ms
t_{pc}	Supply voltages cycle off-to-on time	150	—	—	ms

+1V15 Buck Regulator

+3V3 (16V Max) -> +1V15 / 2A (6A Max)




```
MCU POWER SEQUENCING

STM_ETH_SW_LS_EN  = LOW
STM_1V15_EN       = LOW
STM_ETH_SW_NRESET = LOW       // open-drain: hold reset

wait_until(3V3_PG is HIGH)
STM_ETH_SW_LS_EN = HIGH
wait(T_3V3_ETH_SW_LS_SETTLE)   // monotonic ramps; all rails up within 100 ms

STM_1V15_EN = HIGH
wait_until(BUCK_1V15_PG is HIGH)
wait(T_CLK_START_MAX + 2 ms)  // reset LOW ≥2 ms after rails + clock stable

STM_ETH_SW_NRESET = Hi-Z      // supervisor adds 140–280 ms before release
wait(280 ms)                 // allow maximum delay; rails must remain valid

LAN_init_and_check_ready()


// T_3V3_ETH_SW_LS_SETTLE and T_CLK_START_MAX remain TBD.
```

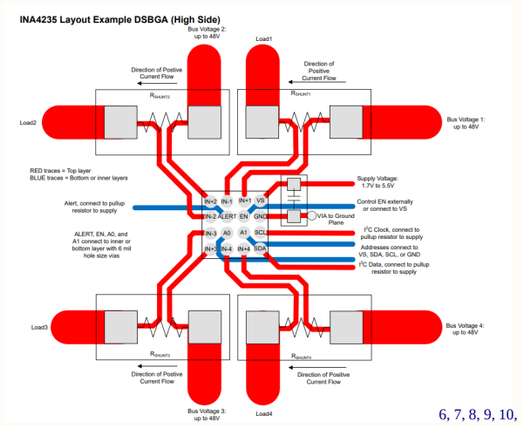
```
Expected maximum load: 1.385 A
= 935mA core/PLL + 400mA analog + 2x25mA SerDes.
Ref: LAN9645xF DS00006065D, Table 5-4, p.191.

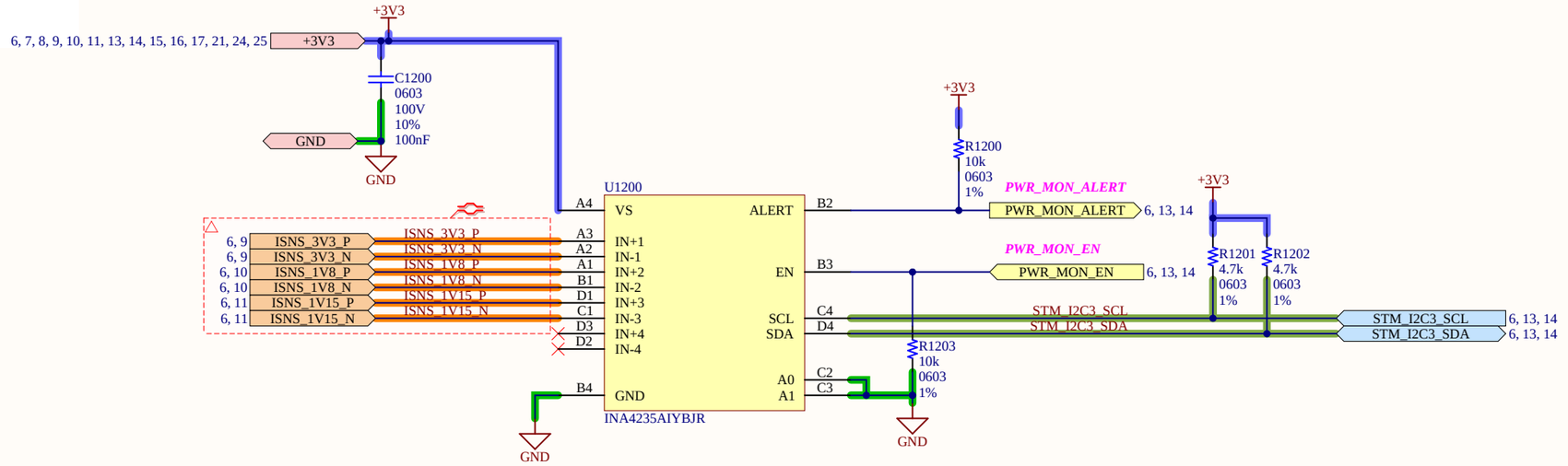
RTRIP=18k: nominal average current-limit onset ≈2.01A,
approximately 45% above expected maximum load.
```

```
1V15 BUCK — NOMINAL CALCULATIONS
VOUT = 0.6*[1+(10.7k+10)/11.5k] = 1.1588 V.
UVP: 80% typ (~0.927V); threshold range 77–83%
OVP: 116% typ (~1.344V); threshold range 113–119%.
OOB: early non-latching correction at 105% typ = ~1.217V.
MODE low: 1.1 MHz FCCM.
At VIN=3.3 V, L=1uH: inductor ripple ≈0.684 A p-p.
RTRIP=18k: valley limit ≈1.667 A;
average current-limit onset ≈2.008 A.
CSS=47nF: soft-start ramp ≈3.13 ms,
excluding startup and PGOOD delays.

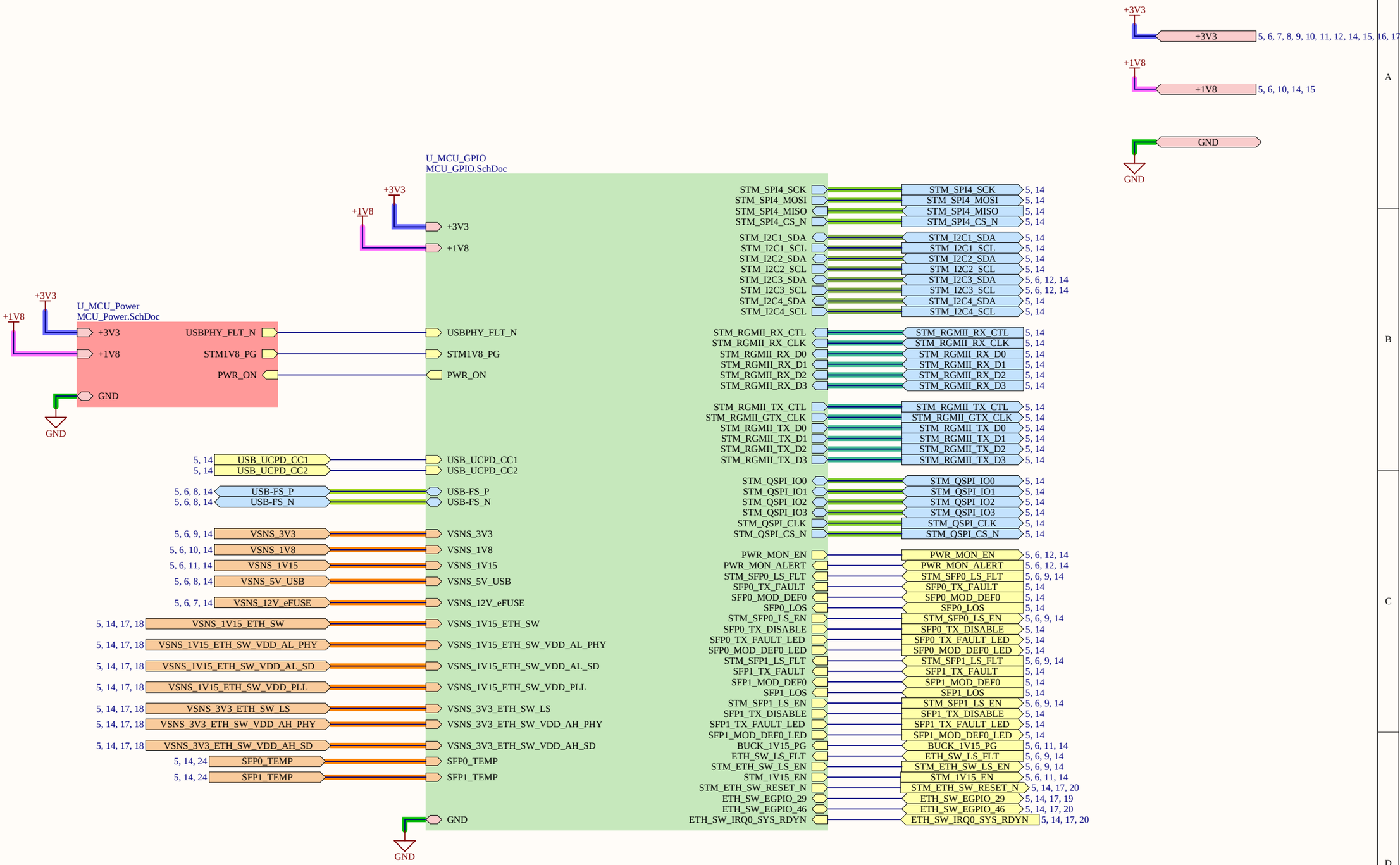
Ref: TI SLVSFQ0B, §§6.3.3–6.3.4 p.12,
Table 6-1 p.13, §6.3.7 p.15.
```

Project: ***Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb***		
Sheet Title: ***Power_1V15.SchDoc***		
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*
Date: 11/5/2026 Revision: 1.0	Sheet: 11 of: 26	



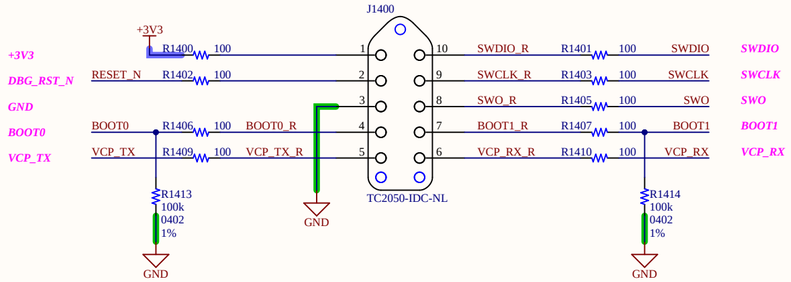


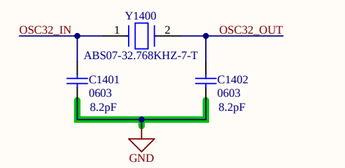
Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*			
Sheet Title: *Power_Monitor.SchDoc*			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	
Date: 11/5/2026	Revision: 1.0	Sheet: 12 of: 26	

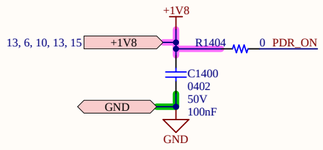


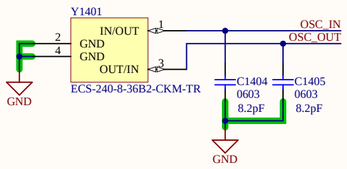
Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*
Sheet Title: *MCU_Top.SchDoc*

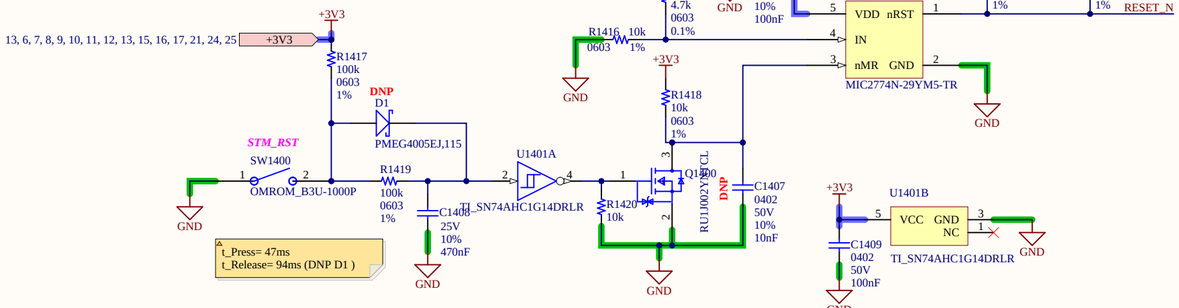
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*
Date: 11/5/2026 Revision: 1.0 Sheet: 13 of: 26		











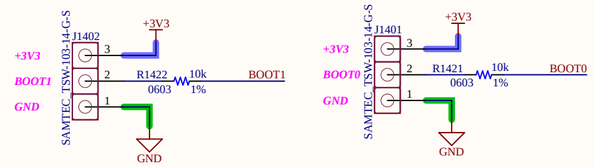
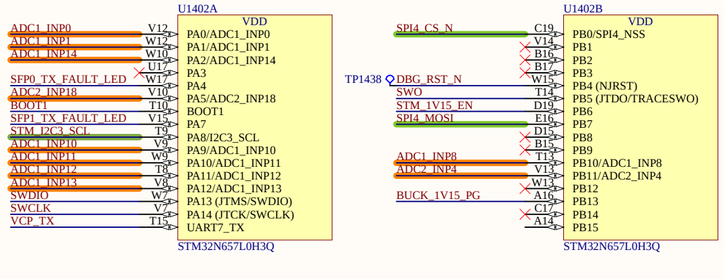
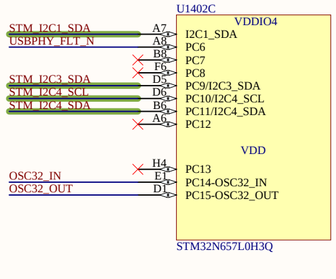
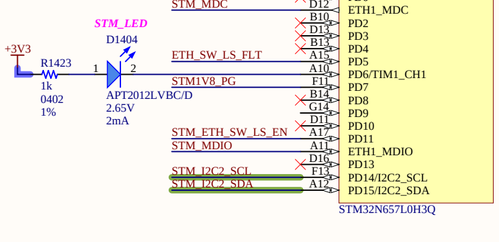


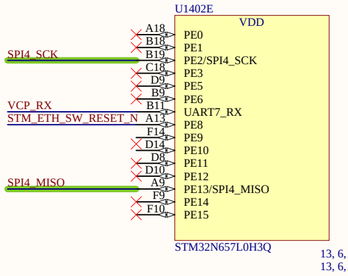
Table 32. Boot modes

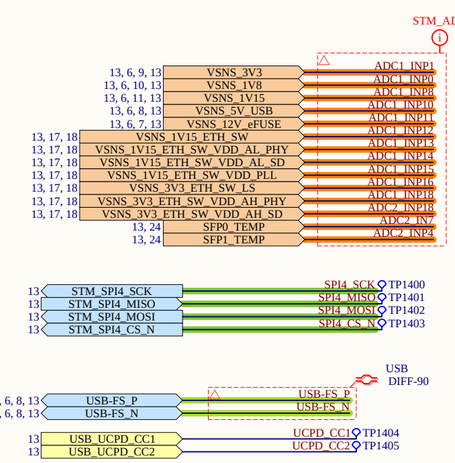
BOOT0	BOOT1	Boot source #1
-	1	Development boot
0	0	Flash boot
1	0	Serial boot

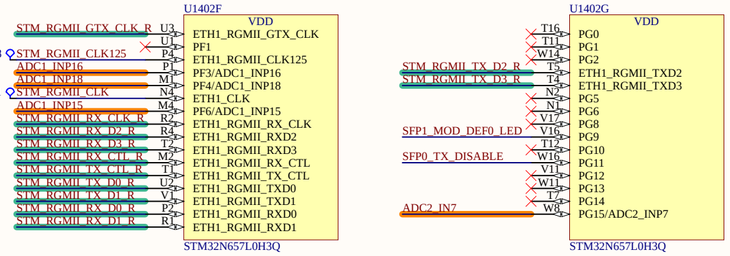


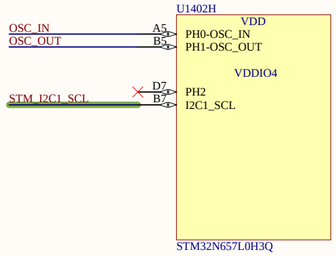


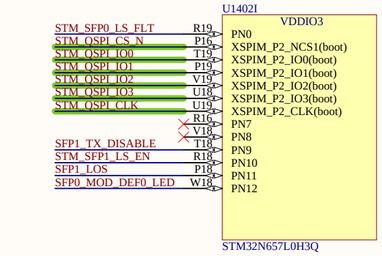


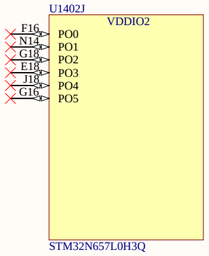


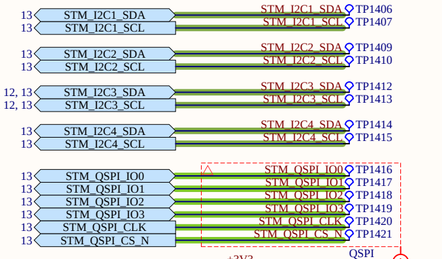


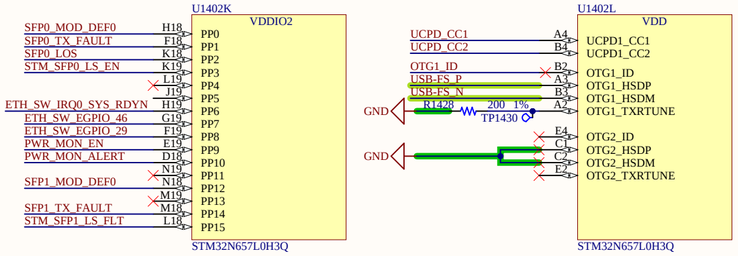


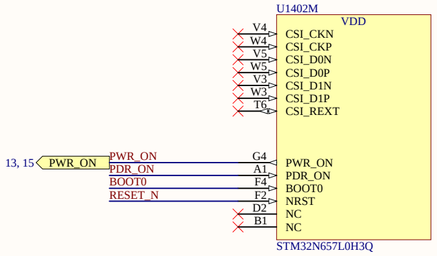


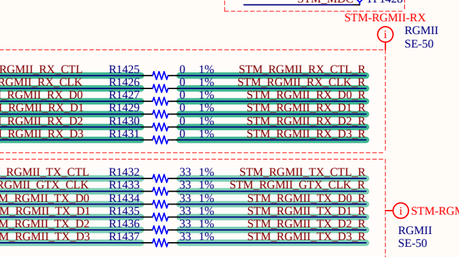




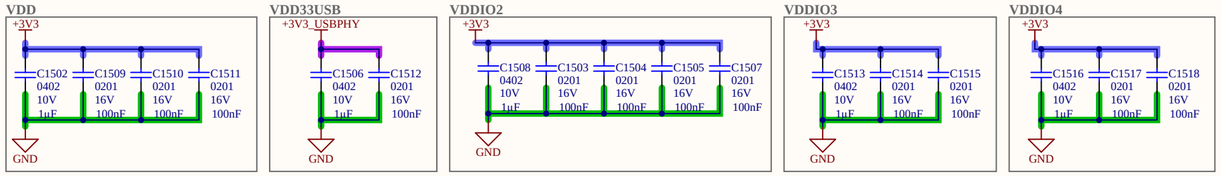


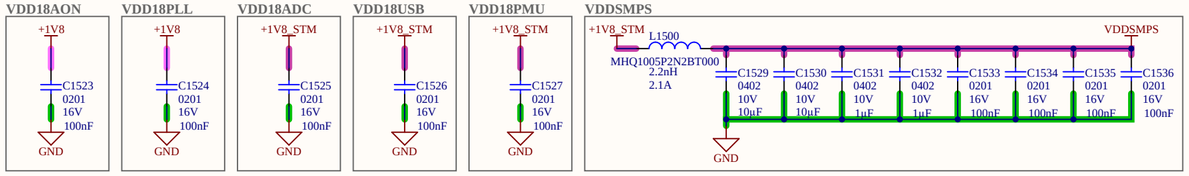


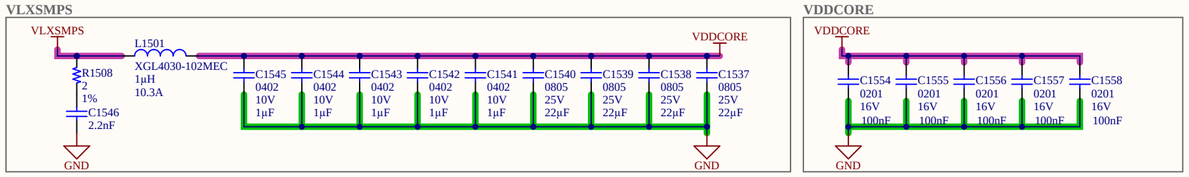


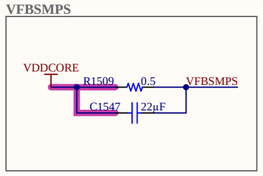


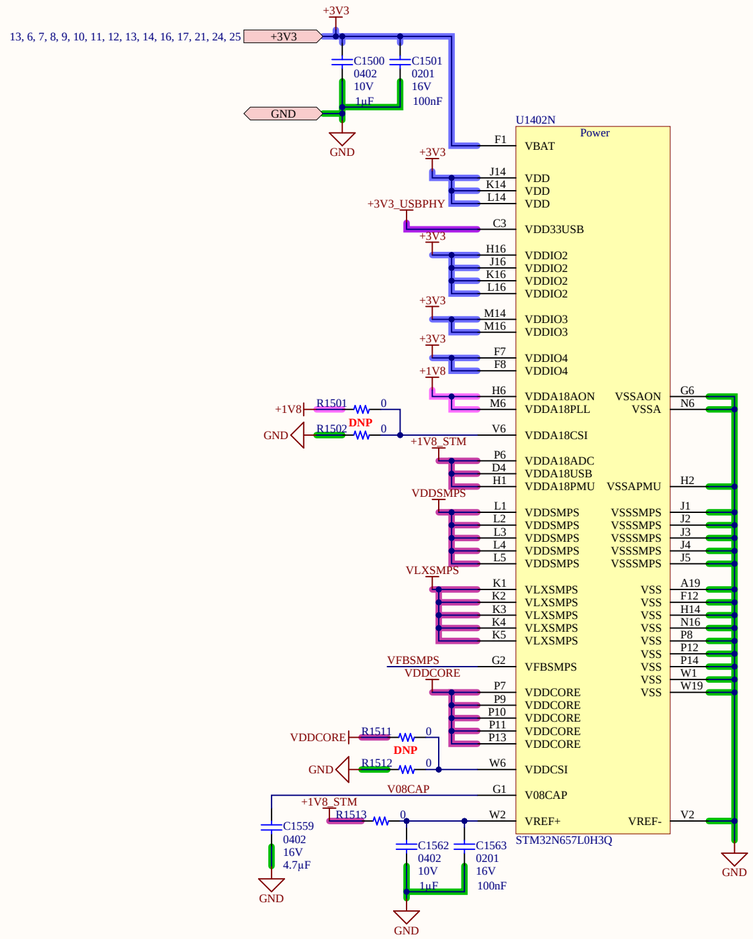
Project:	*Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*
Sheet Title:	*MCU_GPIO.SchDoc*
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi
Date: 11/5/2026	Revision: 1.0 Sheet: 14 of: 26
Yellow Jacket Space Program *Georgia Tech* *Atlanta, GA*	











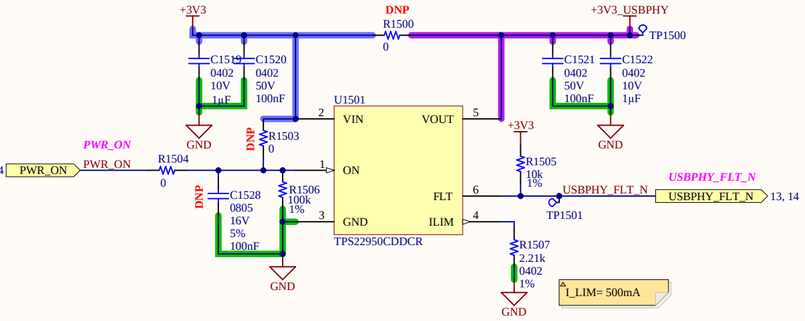


Figure 18. Device startup (V_{CORE} supplied directly from SMPS step-down converter)

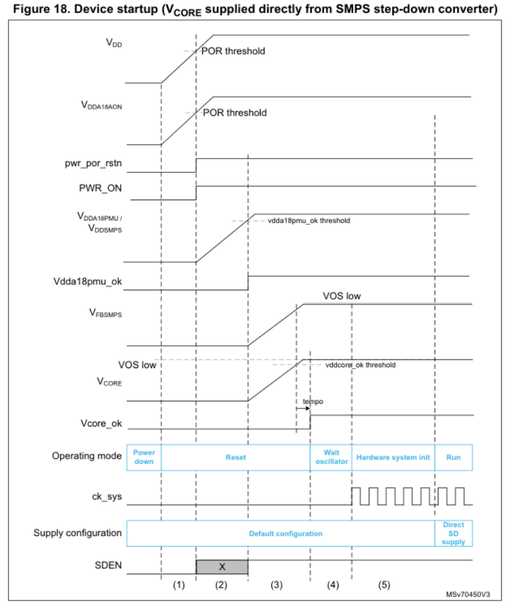
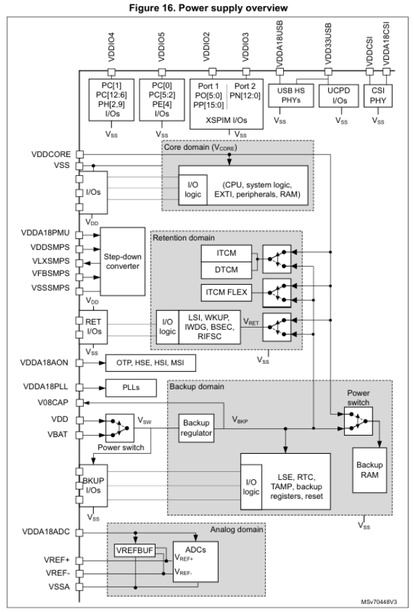


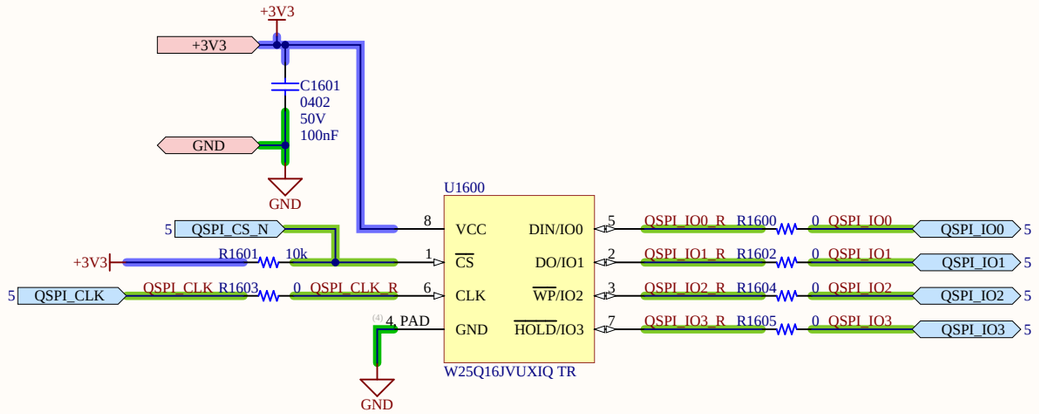
Figure 16. Power supply overview



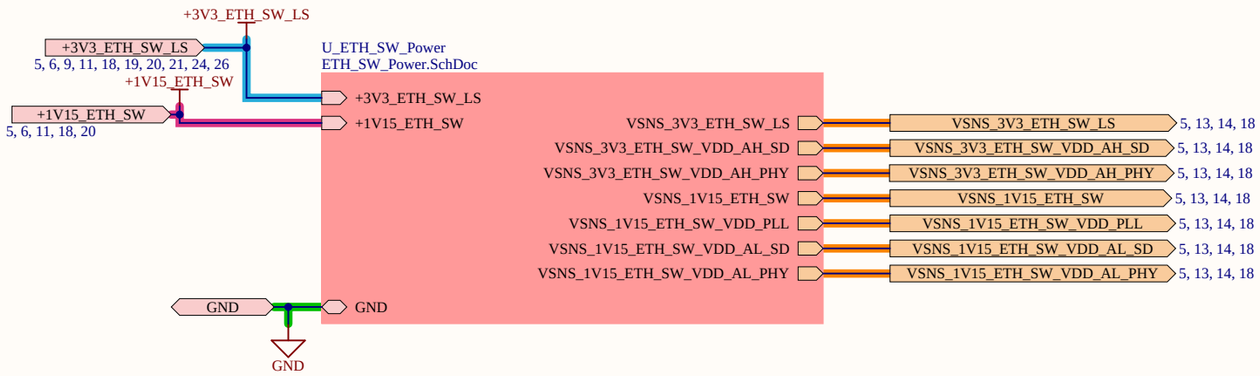
Project: ***Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb***			
Sheet Title: ***MCU_Power.SchDoc***			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	YJSP
Date: 11/5/2026 Revision: 1.0	Sheet: 15 of: 26		

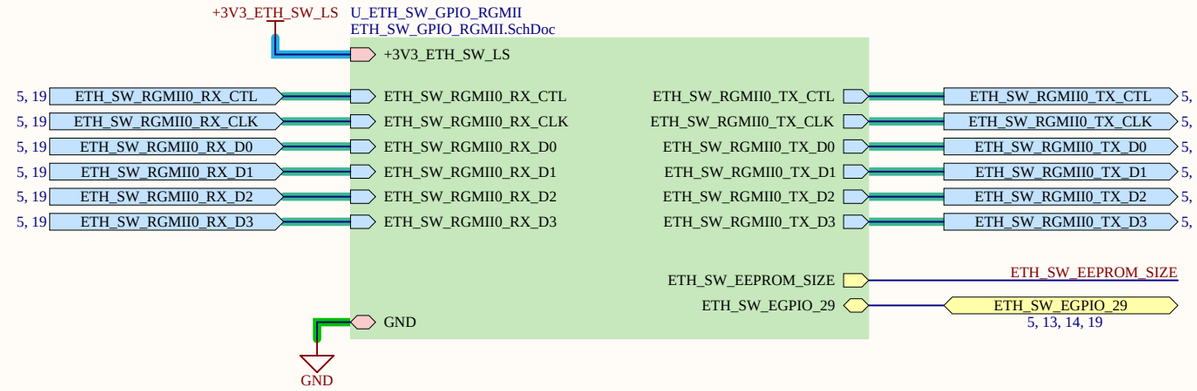
System NOR Flash

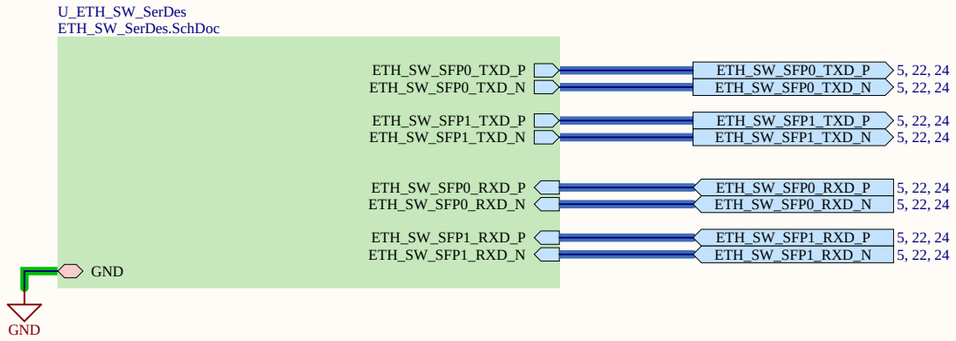
16Mbit 133Mhz QSPI

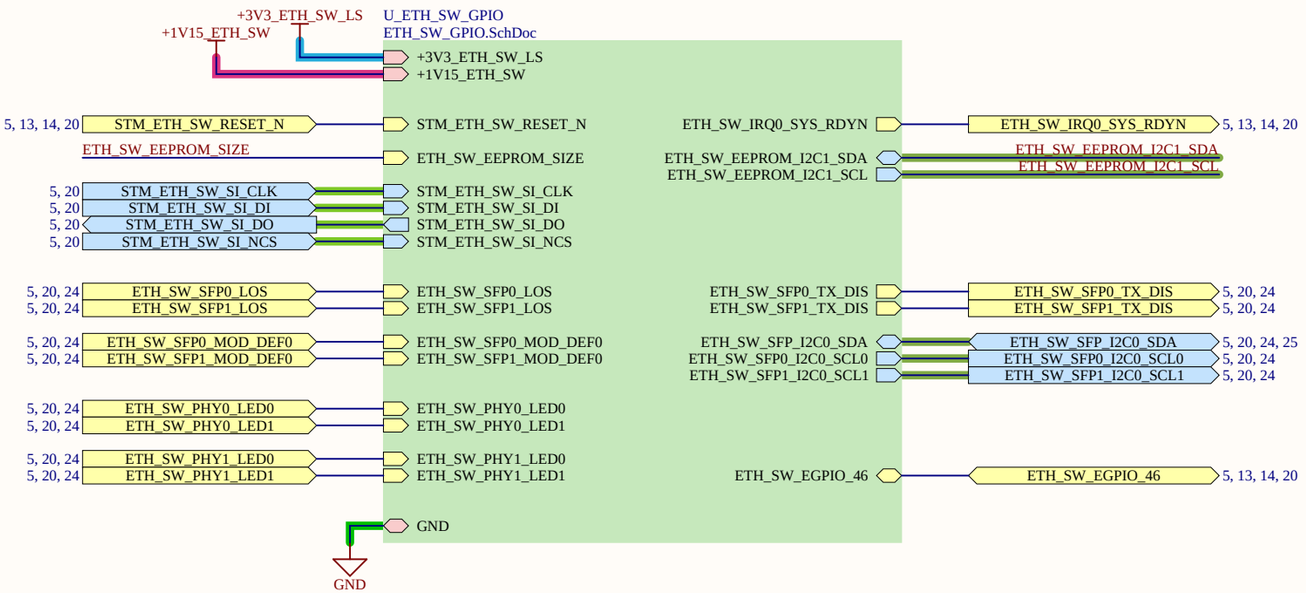


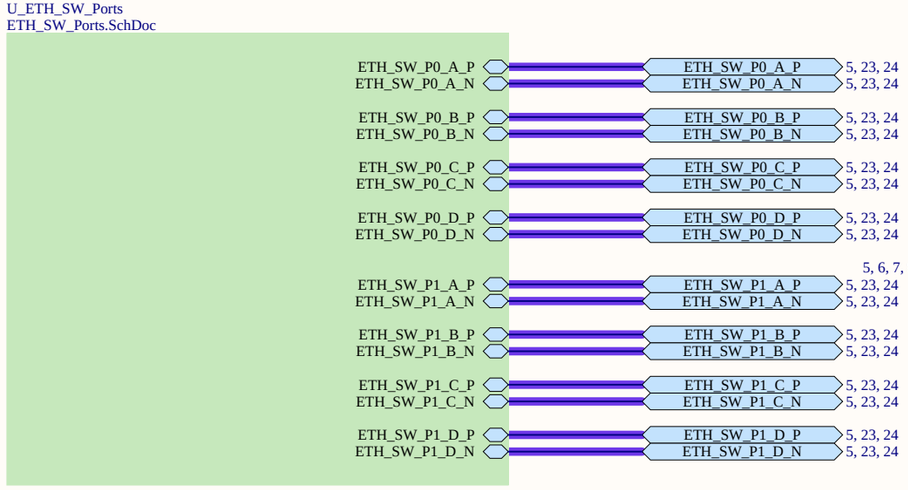
Project: ***Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb***			
Sheet Title: ***Misc.SchDoc***			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	
Date: 11/5/2026 Revision: 1.0	Sheet: 16 of: 26		

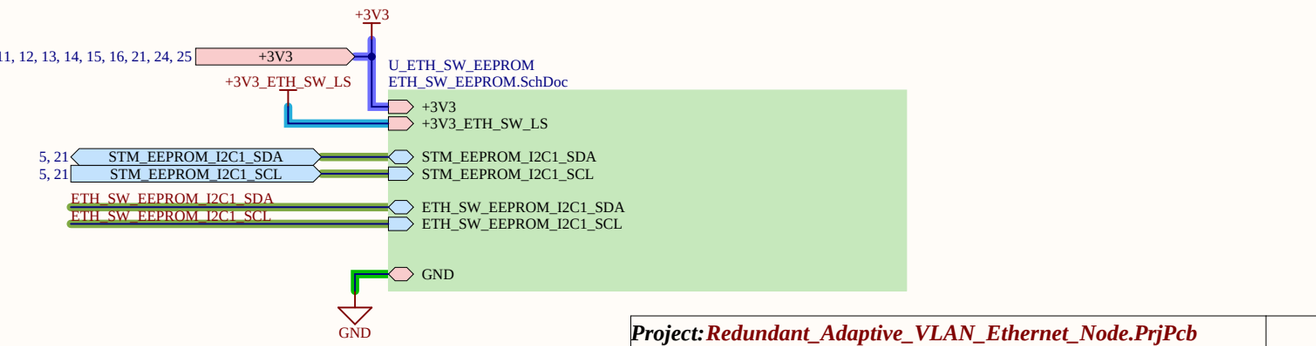












Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*			
Sheet Title: *ETH_SW_Top.SchDoc*			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	YJSP
Date: 11/5/2026 Revision: 1.0	Sheet: 17 of: 26		

VDD — CORE, 1.15 V
Current: 710 mA typ / 935 mA max, INCLUDING VDD_PLL.
Combined core + PLL budget; do not count twice.
14 supply pins; 18 fitted caps (17x100nF + 1x4.7uF);
Cnom=6.40uF; Ceff=6.303uF*

CAPACITORS REFERENCES

https://pim.murata.com/en-us/pim/details/?partNum=GRM033Z71C104ME14%23
https://pim.murata.com/en-us/pim/details/?partNum=GRM188Z71C475KE21%23

FERRITE BEAD REFERENCE

https://pim.murata.com/en-us/pim/details/?partNum=BLM18PG330SN1%23

+1V15_ETH_SW
DNP C1800 0603 16V 10% 4.7µF
C1801 0603 16V 10% 4.7µF
C1802 C1803 C1804 C1805 C1806 C1807 C1808 C1809 C1810 C1811 C1812 C1813 C1814 C1815 C1816 C1817 C1818 0201 16V 20% 100nF
GND

VDD_AL — PHY ANALOG LOW, 1.15 V
Current: 400 mA typ / 400 mA max for the complete group.
12 supply pins; 13 fitted caps (12x100nF + 1x4.7uF);
Cnom=5.90uF; Ceff=5.813uF*

VDD_ALS0 / VDD_ALS1 — SERDES ANALOG LOW, 1.15 V
Current: 25 mA typ / max PER DOMAIN.
Combined current for shared filtered supply: 50 mA max.
2 supply pins; 2 fitted caps (2x100nF);
Cnom=0.20uF; Ceff=0.196uF*

+1V15_ETH_SW_VDD_AL_PHY
DNP C1820 C1821 C1822 C1823 C1824 C1825 C1826 C1827 C1828 C1829 C1830 C1831 C1832 C1833 0603 16V 10% 4.7µF / 0201 16V 20% 100nF
GND

+1V15_ETH_SW_VDD_AL_SD
DNP C1834 0603 16V 10% 4.7µF
C1835 C1836 0201 16V 20% 100nF
GND

VDD_AH — PHY ANALOG HIGH, 3.3 V
Current: 490 mA typ / 505 mA max COMBINED with VDD_AHS.
Separate PHY and SerDes branch currents aren't specified.
6 supply pins; 7 fitted caps (6x100nF + 1x4.7uF);
Cnom=5.30uF; Ceff=4.167uF*

VDD_AHS — SERDES ANALOG HIGH, 3.3 V
Current included in VDD_AH + VDD_AHS 490/505 mA budget.
Do not count the combined budget twice.
2 supply pins; 2 fitted caps (2x100nF);
Cnom=0.20uF; Ceff=0.194uF*

+3V3_ETH_SW_VDD_AH_PHY
DNP C1840 C1841 C1842 C1843 C1844 C1845 C1846 C1847 0603 16V 10% 4.7µF / 0201 16V 20% 100nF
GND

+3V3_ETH_SW_VDD_AH_SD
DNP C1848 0603 16V 10% 4.7µF
C1849 C1850 0201 16V 20% 100nF
GND

VDD_IO_A/B/C/D/E — I/O SUPPLIES, 3.3 V
Combined current budget: 70 mA typ / 80 mA max.
8 supply pins; 9 fitted caps (8x100nF + 1x4.7uF);
Cnom=5.50uF; Ceff=4.361uF*

+3V3_ETH_SW_LS
+3V3_ETH_SW_LS
17, 6, 9, 11, 17, 19, 20, 21, 24, 26
DNP C1853 0603 16V 10% 4.7µF
C1854 0603 16V 10% 4.7µF
C1855 C1856 C1857 C1858 C1859 C1860 C1861 C1862 0201 16V 20% 100nF
GND

DATA SHEET REFERENCES

[1] Microchip LAN9645xF Data Sheet, DS00006065D.
§5.2, p.189: operating supply-voltage requirements.
§5.3, Fig.5-1 / Table 5-1, p.190: power sequencing.
§5.5, Table 5-4 / Note 5-5, p.191: supply-current budgets.

[2] Microchip LAN96459F EDS2 Daughter Card User Guide,
DS50003971A, Fig.A-2, p.31: reference power circuitry,
analog/PLL filtering and supply decoupling.

+1V15_ETH_SW
R1800 1k 0603 1%
Gain= 0.909V/V
VSNS_1V15_ETH_SW 17, 13, 14, 17
R1801 10k 0603 1%
GND

+1V15_ETH_SW_VDD_AL_PHY
R1802 1k 0603 1%
Gain= 0.909V/V
VSNS_1V15_ETH_SW_VDD_AL_PHY 17, 13, 14, 17
R1803 10k 0603 1%
GND

+1V15_ETH_SW_VDD_AL_SD
R1804 1k 0603 1%
Gain= 0.909V/V
VSNS_1V15_ETH_SW_VDD_AL_SD 17, 13, 14, 17
R1805 10k 0603 1%
GND

+1V15_ETH_SW_VDD_PLL
R1807 1k 0603 1%
Gain= 0.909V/V
VSNS_1V15_ETH_SW_VDD_PLL 17, 13, 14, 17
R1808 10k 0603 1%
GND

+1V15_ETH_SW
17, 6, 11, 17, 20 +1V15_ETH_SW
FB1800 3A 33Ω
+1V15_ETH_SW_VDD_AL_PHY
C1819 0201 16V 20% 100nF
GND

+3V3_ETH_SW_LS
100nF 20% C1837 0201 16V
FB1801 3A 33Ω
+3V3_ETH_SW_VDD_AH_PHY
GND

+3V3_ETH_SW_LS
100nF 20% C1838 0201 16V
FB1802 3A 33Ω
+3V3_ETH_SW_VDD_AH_SD
GND

+1V15_ETH_SW
FB1803 3A 33Ω
+1V15_ETH_SW_VDD_AL_SD
C1839 0201 16V 20% 100nF
GND

+1V15_ETH_SW
FB1804 3A 33Ω
+1V15_ETH_SW_VDD_PLL
C1852 0201 16V 20% 100nF
GND

R1806 10 0603 1%
ETH_SW_VDD_PLL_R
C1851 0201 16V 20% 100nF
GND

U1800J
POWER
CORE - 1.15 V
43 VDD
53 VDD
59 VDD
65 VDD
67 VDD
69 VDD
73 VDD
87 VDD
95 VDD
96 VDD
97 VDD
105 VDD
149 VDD
150 VDD
PHY ANALOG LOW - 1.15V
4 VDD_AL
6 VDD_AL
12 VDD_AL
16 VDD_AL
22 VDD_AL
26 VDD_AL
32 VDD_AL
34 VDD_AL
125 VDD_AL
133 VDD_AL
139 VDD_AL
145 VDD_AL
PHY/SERDES ANALOG HIGH - 2.5 V OR 3.3 V
2 VDD_AH
8 VDD_AH
19 VDD_AH
30 VDD_AH
129 VDD_AH
141 VDD_AH
SERDES ANALOG HIGH - 2.5 V OR 3.3 V
55 VDD_AHS
61 VDD_AHS
SERDES ANALOG LOW - 1.15 V
57 VDD_ALS0
63 VDD_ALS1
I/O DOMAIN A - 1.8, 2.5 OR 3.3 V
117 VDD_IO_A
I/O DOMAIN B - 1.8, 2.5 OR 3.3 V
101 VDD_IO_B
109 VDD_IO_B
I/O DOMAIN C - 1.8, 2.5 OR 3.3 V
83 VDD_IO_C
91 VDD_IO_C
I/O DOMAIN D - 3.3 V
71 VDD_IO_D
I/O DOMAIN E - 1.8, 2.5 OR 3.3 V
38 VDD_IO_E
48 VDD_IO_E
PLL
155 VDD_PLL
154 RTN_PLL
GROUND
EP ePAD
NO CONNECT - LEAVE OPEN
127 NC
131 NC
137 NC
147 NC
LAN96455F-I/8NW

+3V3_ETH_SW_LS
R1809 18k 0603 1%
Gain= 0.357V/V
VSNS_3V3_ETH_SW_LS 17, 13, 14, 17
R1812 10k 0603 1%
GND

+3V3_ETH_SW_VDD_AH_PHY
R1810 18k 0603 1%
Gain= 0.357V/V
VSNS_3V3_ETH_SW_VDD_AH_PHY 17, 13, 14, 17
R1813 10k 0603 1%
GND

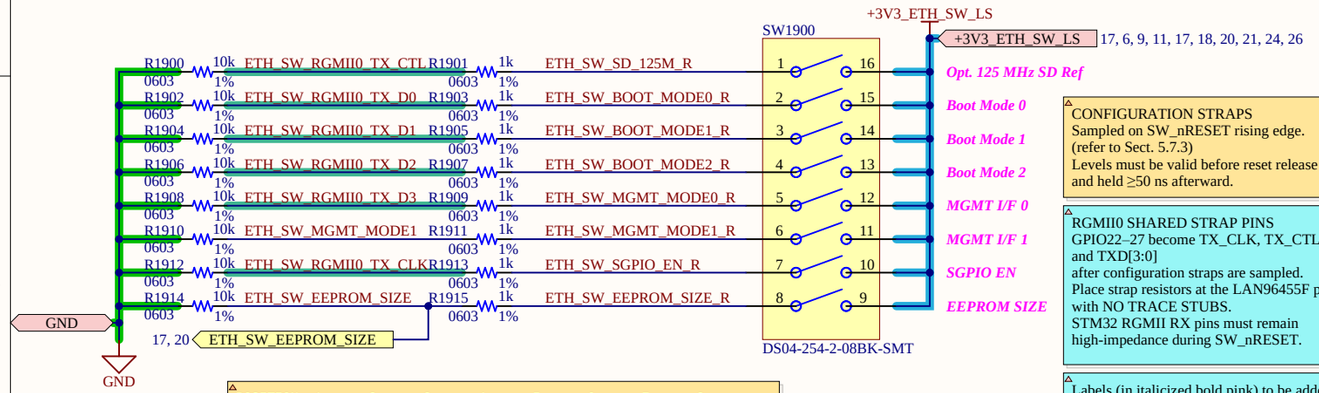
+3V3_ETH_SW_VDD_AH_SD
R1811 18k 0603 1%
Gain= 0.357V/V
VSNS_3V3_ETH_SW_VDD_AH_SD 17, 13, 14, 17
R1814 10k 0603 1%
GND

Project: Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb			
Sheet Title: ETH_SW_Power.SchDoc			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	Yellow Jacket Space Program Georgia Tech Atlanta, GA	
Date: 11/5/2026 Revision: 1.0	Sheet: 18 of: 26		

1	2	3	4	5	6

LAN96455F BOOT AND MANAGEMENT MODES (refer to Sect. 3.4.1)

MGMT M1 M0	BOOT B2 B1 B0	BOOT BEHAVIOR	MANAGEMENT INTERFACE	PROJECT USE
0 0	0 0 0	No boot	I2C client, GPIO[1:0]	Available option
0 0	0 1 1	Internal ROM	I2C client, GPIO[1:0]	Available option
0 0	1 0 0	Internal ROM + I2C table	I2C client, GPIO[1:0]	Available option
1 0	0 0 0	No boot	MIIM client, GPIO[50:48]	Available option
1 0	0 1 1	Internal ROM	MIIM client, GPIO[50:48]	Available option
1 0	1 0 0	Internal ROM + I2C table	MIIM client, GPIO[50:48]	Available option
1 1	0 0 0	No boot	SPI client, GPIO[7:4]	SELECTED BASELINE
1 1	0 1 1	Internal ROM	SPI client, GPIO[7:4]	Available option
1 1	1 0 0	Internal ROM + I2C table	SPI client, GPIO[7:4]	BACKUP CONFIG



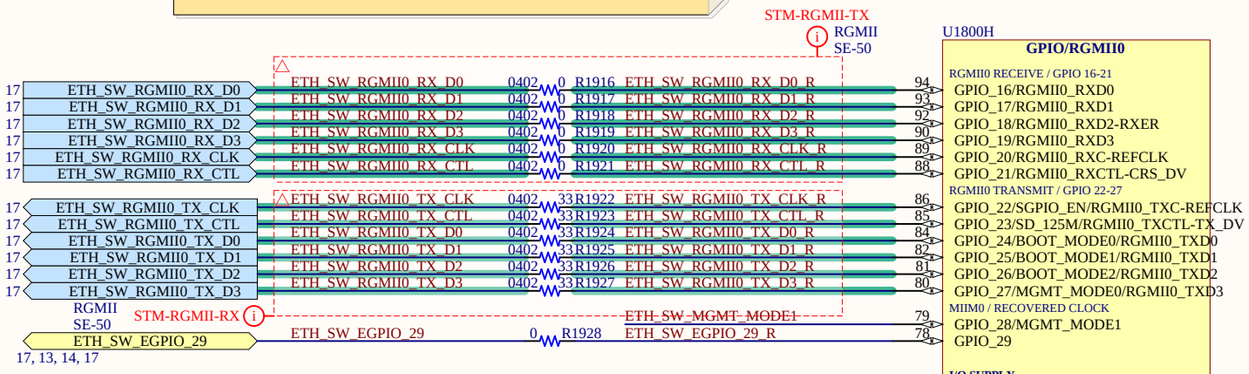
CONFIGURATION STRAPS
Sampled on SW_nRESET rising edge.
(refer to Sect. 5.7.3)
Levels must be valid before reset release and held ≥50 ns afterward.

RGMII0 SHARED STRAP PINS
GPIO22–27 become TX_CLK, TX_CTL, and TXD[3:0]
after configuration straps are sampled.
Place strap resistors at the LAN96455F pins with NO TRACE STUBS.
STM32 RGMII RX pins must remain high-impedance during SW_nRESET.

Labels (in italicized bold pink) to be added as silkscreen on layout

POSITION	1	2	3	4	5	6	7	8
FUNCTION	125M	BM0	BM1	BM2	MG0	MG1	SGPIO	EE_SIZE
DEFAULT	0	0	0	0	1	1	0	0
BACKUP	0	0	0	1	1	1	0	0

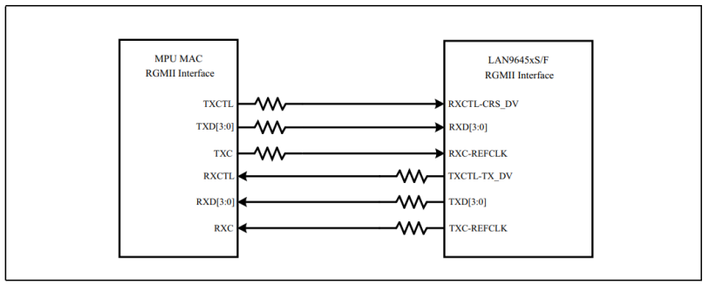
Default: BOOT=000 | Backup: BOOT=100 | Both: SPI management



8.4 RGMII/RMII Connection to other MAC Device

This section describes the recommended methods for connecting the LAN9645xS/F RGMII and RMII interfaces to external MAC devices, such as CPUs, MPUs, or SoCs.

FIGURE 8-5: RGMII MAC CONNECTION



LAN96455F GPIO ASSIGNMENTS - BOOT 000 / MGMT 11
Runtime configuration by STM32 through SPI.

GPIO	MODE	CONNECTION / SIGNAL
0	ALT5	UART_RXD / optional debug header
1	ALT5	UART_TXD / optional debug header
2	ALT3	EEPROM / I2C1_SDA
3	ALT3	EEPROM / I2C1_SCL
4	ALT6	STM32 / SI_SLV_DI (MOSI)
5	ALT6	STM32 / SI_SLV_DO (MISO)
6	ALT6	STM32 / SI_SLV_nCS
7	ALT6	STM32 / SI_SLV_CLK
8	GPIO IN	Optional SFP0 / MOD_DEF0
9	GPIO IN	Optional SFP1 / MOD_DEF0
10-12	GPIO IN	Strap networks; no runtime use
13	ALT5	Backup SFP LED / MAC_LED6
14	ALT5	Primary SFP LED / MAC_LED5
15	GPIO IN	SFP_EN strap; no runtime use
16	ALT1	STM32 / RGMII0_RXD0
17	ALT1	STM32 / RGMII0_RXD1
18	ALT1	STM32 / RGMII0_RXD2
19	ALT1	STM32 / RGMII0_RXD3
20	ALT1	STM32 / RGMII0_RX_CLK
21	ALT1	STM32 / RGMII0_RX_CTL
22	ALT1	STM32 / RGMII0_TX_CLK
23	ALT1	STM32 / RGMII0_TX_CTL
24	ALT1	STM32 / RGMII0_TXD0
25	ALT1	STM32 / RGMII0_TXD1
26	ALT1	STM32 / RGMII0_TXD2
27	ALT1	STM32 / RGMII0_TXD3
28	GPIO IN	MGMT_MODE1 strap; no runtime use
29	GPIO I/O	Spare bidirectional breakout
30	ALT2 (3)	Optional SFP bus / I2C0_SDA
31	ALT2 (3)	Optional SFP0 / I2C0_SCL0
32	ALT2 (3)	Optional SFP1 / I2C0_SCL1
33	ALT2 (4)	Optional STM status / IRQ0
34-41	GPIO IN	Strap networks; no runtime use
42	ALT6 (3)	Optional SFP0 / SD0 from LOS
43	ALT6 (3)	Optional SFP1 / SD1 from LOS
E44	GPIO (3)	Optional SFP0 / TX_DISABLE
E45	GPIO (3)	Optional SFP1 / TX_DISABLE
E46	GPIO I/O	Spare bidirectional breakout
E47	ALT2	Debug RJ45 / PHY1_LED0
E48	ALT2	Debug RJ45 / PHY1_LED1
E49	ALT2	Laptop RJ45 / PHY0_LED0
E50	ALT2	Laptop RJ45 / PHY0_LED1

1. SPI client uses GPIO0-3, ALT2, VDD_IO_A.
   Fit the four STM32-to-GPIO0-3 links.
   Keep R2034-R2037 DNP (old GPIO4-7 SPI paths).
   DI = STM32 MOSI; DO = STM32 MISO.
2. Keep UART links R2052/R2053 DNP.
   Keep EEPROM links R2054/R2055 DNP.
   GPIO0-3 are reserved for SPI in this assembly.
   BOOT000 does not start internal firmware.
3. Enable only for selected/populated LAN sideband option.
   Otherwise leave unused pins as inputs.
   E44/E45: output only when LAN owns TX_DISABLE;
   initialize HIGH (disabled) before enabling output.
   Verify LOS polarity and SFP I2C ownership.
4. Proposed managed-mode IRQ use; configure interrupt sources. This is not SYS_RDYn in BOOT000.

GPIO IN/OUT are ordinary GPIO directions, not ALT numbers.
Retain all required reset-time strap networks.
Unmanaged straps do not configure BOOT000 runtime behavior.
Do not drive switch inputs while switch power is OFF.

REF: LAN9645xF DS00006065D:
Table 3-4 pp.28-29; Table 3-5 pp.30-31;
Sections 3.4.1.1-2 pp.42-43; Section 5.2 p.189.

LAN96455F GPIO TARGET - BOOT 100 / EEPROM BACKUP
ROM + I2C EEPROM; STM32 SPI management disconnected.
BACKUP ONLY: straps/EEPROM/ROM operation not yet validated.

GPIO	TARGET	CONNECTION / SIGNAL	ACTION
0	ALT5 *	Optional debug header / UART_RXD	fix DNP (1)
1	ALT5 *	Optional debug header / UART_TXD	fix DNP (1)
2	ALT3	Boot EEPROM / I2C1_SDA	fix DNP (2)
3	ALT3	Boot EEPROM / I2C1_SCL	fix DNP (2)
4	TBD *	STM32 / SI_SLV_DI path disconnected	fix DNP (3)
5	TBD *	STM32 / SI_SLV_DO path disconnected	fix DNP (3)
6	TBD *	STM32 / SI_SLV_nCS path disconnected	fix DNP (3)
7	TBD *	STM32 / SI_SLV_CLK path disconnected	fix DNP (3)
8	GPIO IN *	Optional SFP0 / MOD_DEF0	fix DNP (4)
9	GPIO IN *	Optional SFP1 / MOD_DEF0	fix DNP (4)
10-12	IN *	Strap networks; no intended runtime use	
13	ALT5 *	Backup SFP LED / MAC_LED6	
14	ALT5 *	Primary SFP LED / MAC_LED5	
15	IN *	SFP_EN strap; no intended runtime use	
16	ALT1 *	STM32 / RGMII0_RXD0	
17	ALT1 *	STM32 / RGMII0_RXD1	
18	ALT1 *	STM32 / RGMII0_RXD2	
19	ALT1 *	STM32 / RGMII0_RXD3	
20	ALT1 *	STM32 / RGMII0_RX_CLK	
21	ALT1 *	STM32 / RGMII0_RX_CTL	
22	ALT1 *	STM32 / RGMII0_TX_CLK	
23	ALT1 *	STM32 / RGMII0_TX_CTL	
24	ALT1 *	STM32 / RGMII0_TXD0	
25	ALT1 *	STM32 / RGMII0_TXD1	
26	ALT1 *	STM32 / RGMII0_TXD2	
27	ALT1 *	STM32 / RGMII0_TXD3	
28	IN *	MGMT_MODE1 strap; backup setting TBD	
29	GPIO I/O *	Spare bidirectional breakout	
30	ALT2 *	Optional SFP bus / I2C0_SDA	fix DNP (4)
31	ALT2 *	Optional SFP0 / I2C0_SCL0	fix DNP (4)
32	ALT2 *	Optional SFP1 / I2C0_SCL1	fix DNP (4)
33	ROM	SYS_RDYn / EEPROM_SIZE strap	
34-41	IN *	Strap networks; no intended runtime use	
42	ALT6 *	Optional SFP0 / SD0 from LOS	fix DNP (4)
43	ALT6 *	Optional SFP1 / SD1 from LOS	fix DNP (4)
E44	GPIO OUT *	Optional SFP0 / TX_DISABLE	fix DNP (4)
E45	GPIO OUT *	Optional SFP1 / TX_DISABLE	fix DNP (4)
E46	GPIO I/O *	Spare bidirectional breakout	
E47	ALT2 *	Debug RJ45 / PHY1_LED0	
E48	ALT2 *	Debug RJ45 / PHY1_LED1	
E49	ALT2 *	Laptop RJ45 / PHY0_LED0	
E50	ALT2 *	Laptop RJ45 / PHY0_LED1	

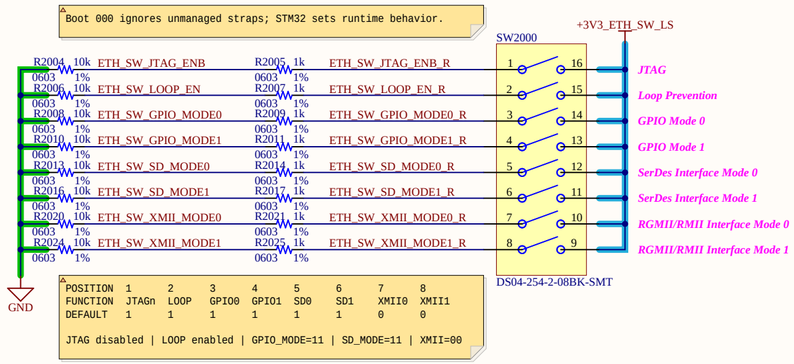
1. GPIO0/1: remove STM32 SPI DI/DO links.
   Keep UART R2052/R2053 DNP unless UART use is validated;
   populate only for the selected UART debug option.
2. GPIO2/3: remove STM32 SPI nCS/CLK links.
   Populate EEPROM links R2054/R2055.
   Fit/program EEPROM and required support components.
3. Keep old GPIO4-7 SPI links R2034-R2037 DNP.
   Do not run concurrent STM32 SPI configuration.
4. Populate only selected, validated LAN sideband paths.
   Fit both endpoint links where required.
   Select one SFP I2C controller and one TX_DISABLE owner.
   Keep conflicting STM32 branches disconnected.
   Leave unselected optional paths DNP.

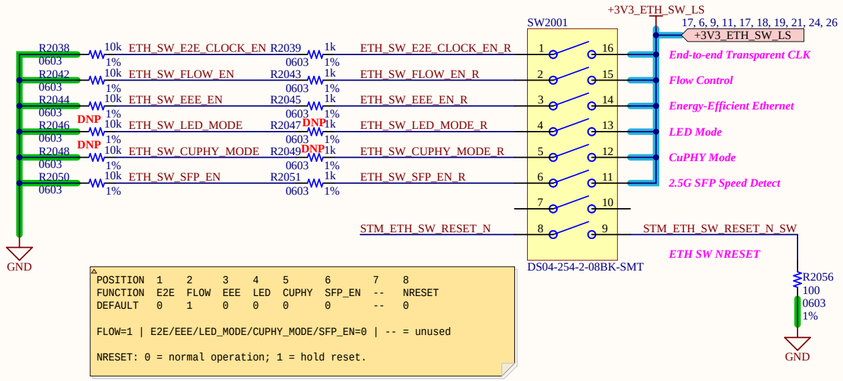
* Proposed target; ROM/EEPROM support must be verified.

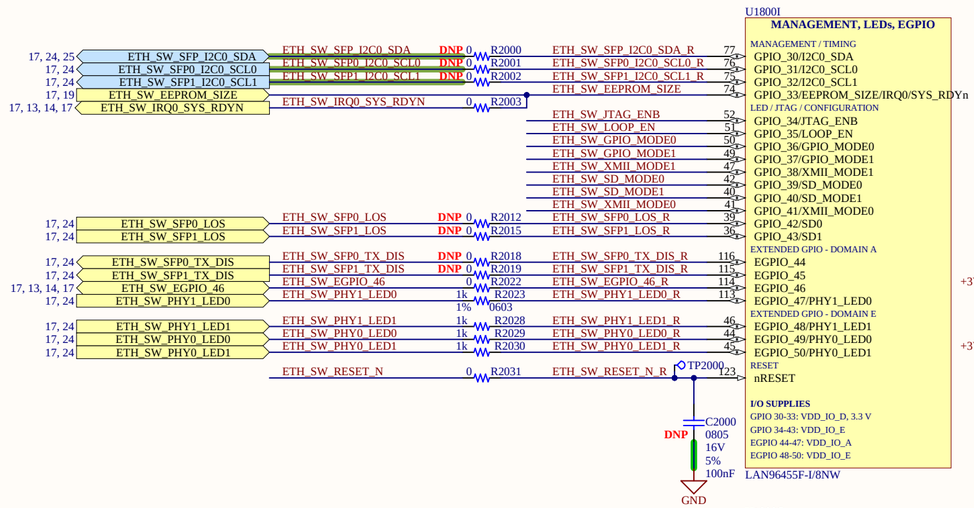
REF: LAN9645xF DS00006065D:
Table 3-5 pp.30-31: hardware ALT functions.
Table 3-9 pp.35-36: stock Mode D assignments.
Table 3-11 p.41: SYS_RDYn and SFP functions.
Sections 3.4.1.1-2 pp.42-43: boot/management selection.
Table 3-12 p.43: boot configuration precedence.
Sec.4.1.2.15 / Table 4-2 pp.51-52: EEPROM overrides.
Supplied Vishnu review, Sec.1: avoid EEPROM/host coexistence.

Project: Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb		
Sheet Title: ETH_SW_GPIO_RGMII.SchDoc		
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	Yellow Jacket Space Program Georgia Tech Atlanta, GA
Date: 11/5/2026 Revision: 1.0	Sheet: 19 of: 26	

1	2	3	4	5	6





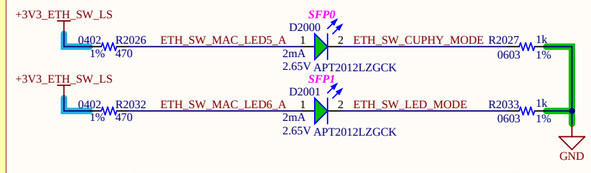


ETHERNET LED STATUS — INTENDED BEHAVIOR

PORT	LED	SIGNAL	OFF	ON	BLINK
Laptop RJ45	Green	PHY0_LED1	No link	Link	—
Laptop RJ45	Yellow	PHY0_LED0	No activity	—	RX/TX activity
Debug RJ45	Green	PHY1_LED1	No link	Link	—
Debug RJ45	Yellow	PHY1_LED0	No activity	—	RX/TX activity
Primary SFP0	Green	MAC_LED5	TBD	TBD	TBD
Backup SFP1	Green	MAC_LED6	TBD	TBD	TBD

No copper link: both associated RJ45 LEDs OFF.

Ref: LAN9645xF DS00006065D, Table 3-11 p.41,
Table 3-12 p.43, Section 3.4.2.5/Table 3-14 p.45.



MAC LEDs — OPTICAL PORT STATUS

MAC5 = primary: GPIO14/pin99; MAC6 = backup: GPIO13/pin100.
Managed: ALT5; supply: VDD_IO_B = +3V3_ETH_SW_LS.
Each: KT-0805G + 470R series + 1k pulldown (1%).
LOW = lit; HIGH = off. HIGH sources ~3.3mA into pulldown;
verify/configure Drive Strength >=4mA drive.

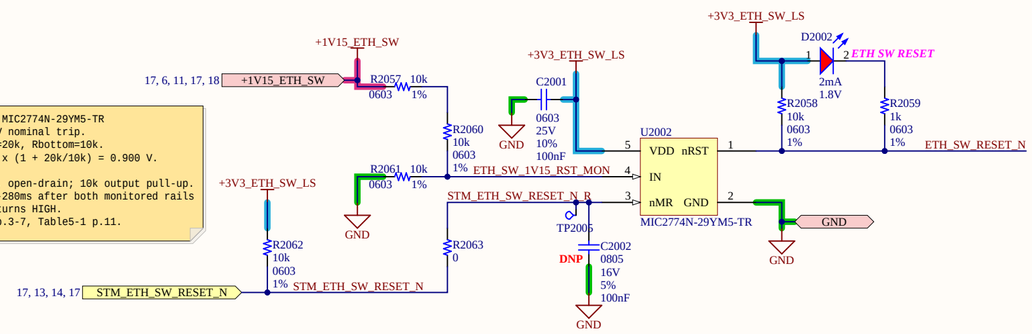
LOW-STRAP CHECK — LED_MODE=0, CUPHY_MODE=0:
At 3.465V (+5% assumed), VIL(max)=0.39VDD=1.351V.
Including resistor tolerance, internal pull-up and leakage:
Vpin ~0.63V assuming LED Vf=2.6V — estimated PASS.
Guaranteed pass requires Vf >=1.513V at ~1.29mA over
temperature

Refs: LAN9645xF DS00006065D, Tables 3-4/3-5 pp.28–30,
Table 5-7 pp.192–194; DS00006273C Fig.4-1 p.9.
KT-0805G A3, Electrical/Optical Characteristics p.3.

HIGH sources ~3.3mA into pulldown;
verify/configure Drive Strength >=4mA drive.

Drive Strength configurability TBD.

ETH SW RESET



Optional UART Debug

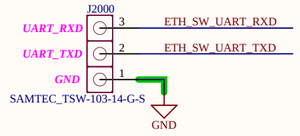
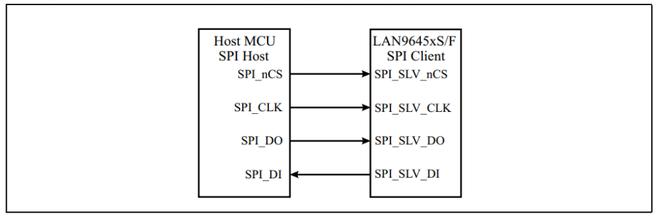


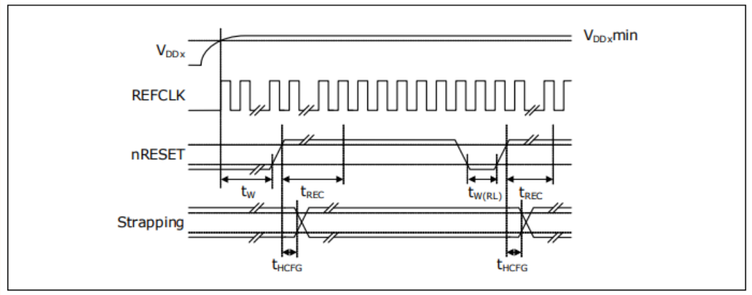
FIGURE 9-1: SPI CONNECTIONS BETWEEN MCU AND LAN9645XS/F



5.7.3 NRESET TIMING SPECIFICATIONS

The nRESET signal waveform and the required measurement points for the timing specification are shown in Figure 5-2.

FIGURE 5-2: NRESET TIMING DIAGRAM



The signal applied to the nRESET input must comply with the specifications listed in the following table at the reset pin of the devices.

TABLE 5-17: NRESET TIMING SPECIFICATIONS

Parameter	Symbol	Min	Typ	Max	Units	Notes
nRESET assertion time after power supplies and clock stabilizes	t_W	2			ms	
nRESET pulse width	$t_{W(RL)}$	1			us	
Hold time for GPIO-mapped strapping pins relative to nRESET	t_{HCFG}	50			ns	
Recovery time from reset inactive to device active (starts reading configuration PROM)	t_{REC}			100	ms	
Recovery time from reset inactive to device fully active (SYSTEM READY signal active in unmanaged device)			500		ns	

TABLE 4-4: LAN96455 PORT CONFIGURATIONS (5 PORTS)

LAN96455 Port Configuration	CuPHY					SGMII		RGMII	
	#0	#1	#2	#3	#4	#0	#1	#0	#1
5x CuPHY	DEV0	DEV1	DEV2	DEV3	DEV4	-	-	-	-
4x CuPHY + 1x 2.5G SGMII	DEV0	DEV1	DEV2	DEV3	-	DEV5	-	-	-
4x CuPHY + 1x RGMII	DEV0	DEV1	DEV2	DEV3	-	-	-	DEV7	-
3x CuPHY + 2x 2.5G SGMII	DEV0	DEV1	DEV2	-	-	DEV5	DEV6	-	-
3x CuPHY + 2x RGMII	DEV0	DEV1	DEV2	-	-	-	-	DEV7	DEV8
3x CuPHY + 1x 2.5G SGMII + 1x RGMII	DEV0	DEV1	DEV2	-	-	DEV5	-	DEV7	-
2x CuPHY + 1x 2.5G SGMII + 2x RGMII	DEV0	DEV1	-	-	-	DEV5	-	DEV7	DEV8
2x CuPHY + 2x 2.5G SGMII + 1x RGMII	DEV0	DEV1	-	-	-	DEV5	DEV6	DEV7	-
1x CuPHY + 2x 2.5G SGMII + 2x RGMII	DEV0	-	-	-	-	DEV5	DEV6	DEV7	DEV8

Project: Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb			
Sheet Title: ETH_SW_GPIO.SchDoc			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	Yellow Jacket Space Program Georgia Tech Atlanta, GA	
Date: 11/5/2026 Revision: 1.0	Sheet: 20 of: 26		

```
24CSM01 I2C PULL-UP RESISTOR LIMITS
Assumptions: VCC = 3.3 V; CL = 100 pF per line
             VOL(max) = 0.4 V; IOL = 2.1 mA

MODE         MAX CLOCK   tr(max)   Rmin       Rmax
-----------  ----------  --------  ---------  ----------
Standard     100 kHz     1000 ns   1.381 kohm 11.802 kohm
Fast         400 kHz      300 ns   1.381 kohm  3.541 kohm
Fast Plus    1.0 MHz      120 ns   1.381 kohm  1.416 kohm
Hs (SDA)     3.4 MHz       80 ns   1.381 kohm  0.944 kohm
```

```
Rmin = (VCC - VOL(max)) / IOL
Rmax = tr(max) / (0.8473 * CL)

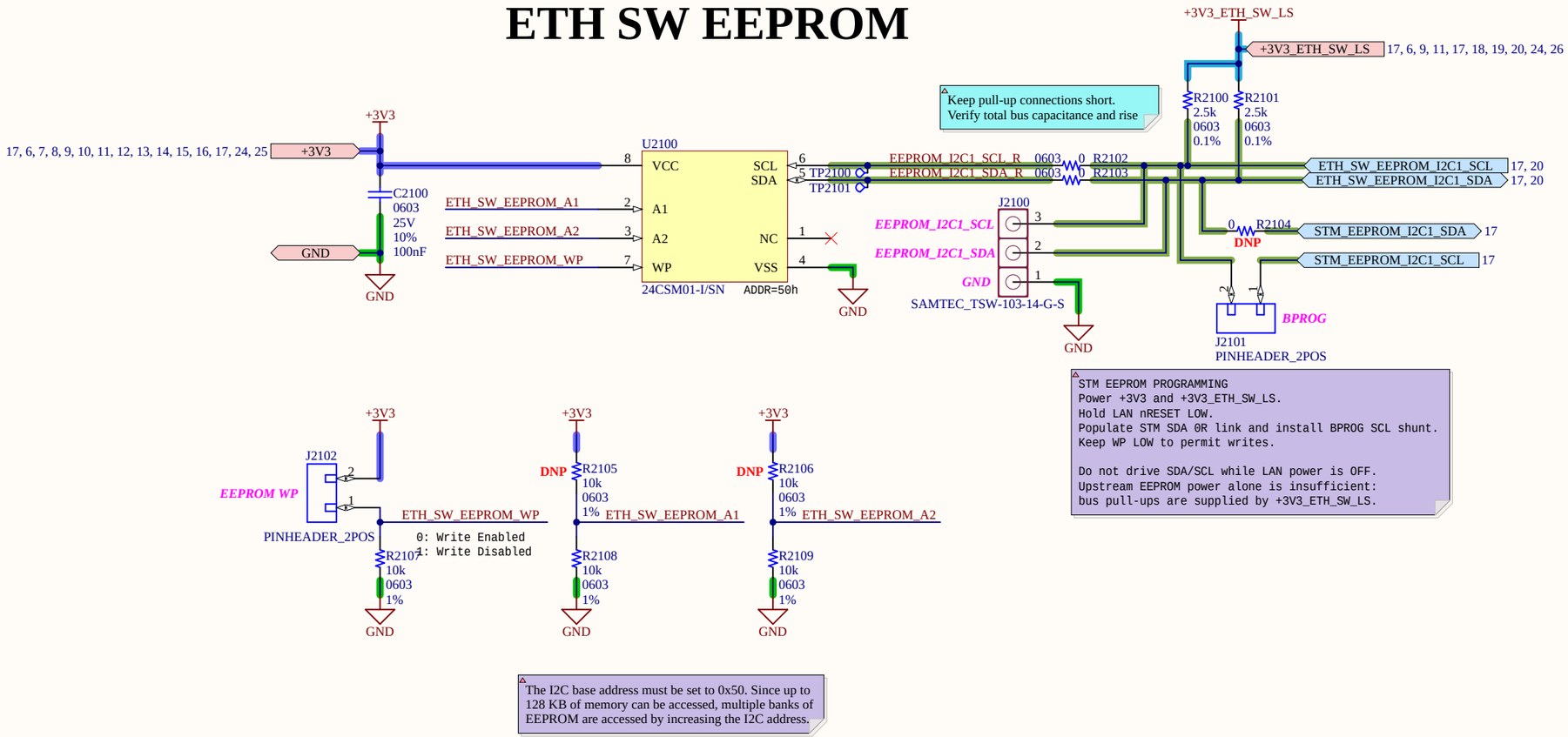
REFERENCES:
Microchip DS20006781D, Rev. D:
  System-configuration diagram, p. 2: equations.
  Table 1-1, D3, p. 4: VOL/IOL guarantee.
NXP UM10204, Rev. 7.0:
  Table 11, p. 44: Standard/Fast/Fast Plus timing.
  Table 13, p. 47: Hs timing at 100 pF.
  Sec. 5.3.1, pp. 34-35: Hs current-source operation.
```

```
tr = 0.8473 * RPU * CL
   = 0.8473 * 2500 * 100e-12
   = 212 ns < 300 ns Fast-mode limit.

IOL = (VPU - VOL(max)) / RPU
    = (3.3 - 0.4) / 2500
    = 1.16 mA < 2.1 mA EEPROM guarantee.

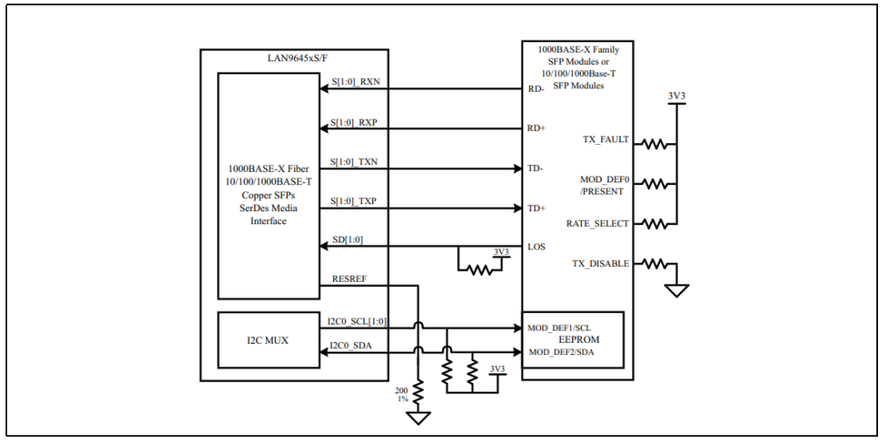
Microchip LAN9645xF, DS00006065D:
  Sec. 4.1.2.15, p. 52:
EEPROM >= 400 kHz requirement.
```

ETH SW EEPROM



Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*			
Sheet Title: *ETH_SW_EEPROM.SchDoc*			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	YJSP
Date: 11/5/2026 Revision: 1.0	Sheet: 21 of: 26		

FIGURE 7-3: SERDES MEDIA INTERFACE TO 1000BASE-X FAMILY FIBER SFP OR 10/100/1000BASE-T COPPER SFP REFERENCE CONNECTION

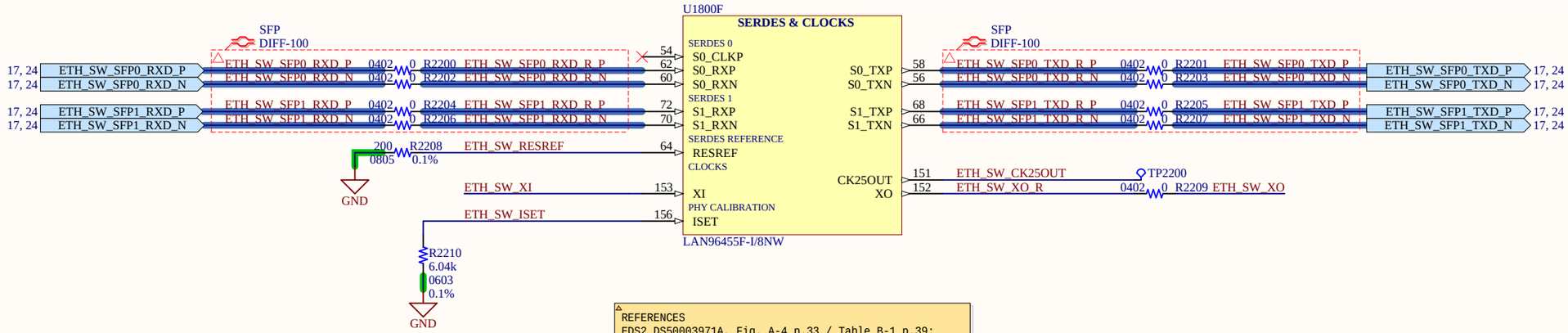


I. Pin Descriptions

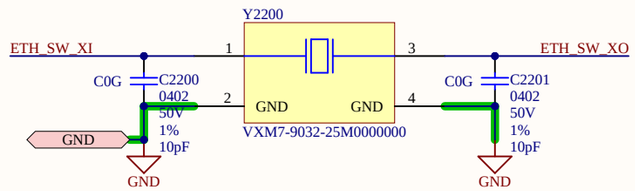
Pin	Symbol	Name/Description	Ref.
1	V_{EET}	Transmitter Ground (Common with Receiver Ground)	1
2	T_{FAULT}	Transmitter Fault.	
3	T_{DIS}	Transmitter Disable. Laser output disabled on high or open.	2
4	MOD_DEF(2)	Module Definition 2. Data line for Serial ID.	3
5	MOD_DEF(1)	Module Definition 1. Clock line for Serial ID.	3
6	MOD_DEF(0)	Module Definition 0. Grounded within the module.	3
7	Rate Select	No connection required	
8	LOS	Loss of Signal indication. Logic 0 indicates normal operation.	4
9	V_{EER}	Receiver Ground (Common with Transmitter Ground)	1
10	V_{EER}	Receiver Ground (Common with Transmitter Ground)	1
11	V_{EER}	Receiver Ground (Common with Transmitter Ground)	1
12	RD-	Receiver Inverted DATA out. AC Coupled	
13	RD+	Receiver Non-inverted DATA out. AC Coupled	
14	V_{EER}	Receiver Ground (Common with Transmitter Ground)	1
15	V_{CCR}	Receiver Power Supply	
16	V_{CCT}	Transmitter Power Supply	
17	V_{EET}	Transmitter Ground (Common with Receiver Ground)	1
18	TD+	Transmitter Non-Inverted DATA in. AC Coupled.	
19	TD-	Transmitter Inverted DATA in. AC Coupled.	
20	V_{EET}	Transmitter Ground (Common with Receiver Ground)	1

UNUSED CLOCK PINS
S0_CLKP (54): NC; SD_125M=0, 25 MHz crystal mode.
CK25OUT (151): 25 MHz clock monitor test point.
Ref: DS00006065D, §4.22 p.163; Table 3-3 p.25.

CALIBRATION RESISTORS
RESREF (pin 64): 200 Ω ±1% to GND.
ISET (pin 156): 6.00 kΩ ±1% to GND (only 6.04 kΩ are available).
Ref: LAN9645xF DS00006065D, §3.2, Table 3-3, pp.23–24; pin numbers: Table 3-2, p.22.



REFERENCES
EDS2 DS50003971A, Fig. A-4 p.33 / Table B-1 p.39: reference capacitor values and crystal selection.
VXM7 rev.30-Apr-2020, Table 2 p.1: pin connections.
LAN DS00006065D, sec.5.8.1 / Table 5-28 p.209: crystal electrical requirements.



25 MHz CRYSTAL REFERENCE
Y: VXM7-9032-25M0000000; specified load CL = 10 pF.
Ca = Cb = 10 pF, C0G/NP0, 1%, 50 V, 0402.

LAYOUT / VALIDATION (PROJECT GUIDANCE)
Place crystal and load capacitors close to XI/XO.
Keep traces short and ground returns local.
Avoid long branches or test-point stubs on XI/XO.
10 pF values follow the reference design; verify frequency and startup on the assembled PCB.
Effective IC/PCB parasitic capacitance remains TBD.

Project: ***Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb***			
Sheet Title: ***ETH_SW_SERDES.SchDoc***			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	
Date: 11/5/2026 Revision: 1.0	Sheet: 22 of: 26		

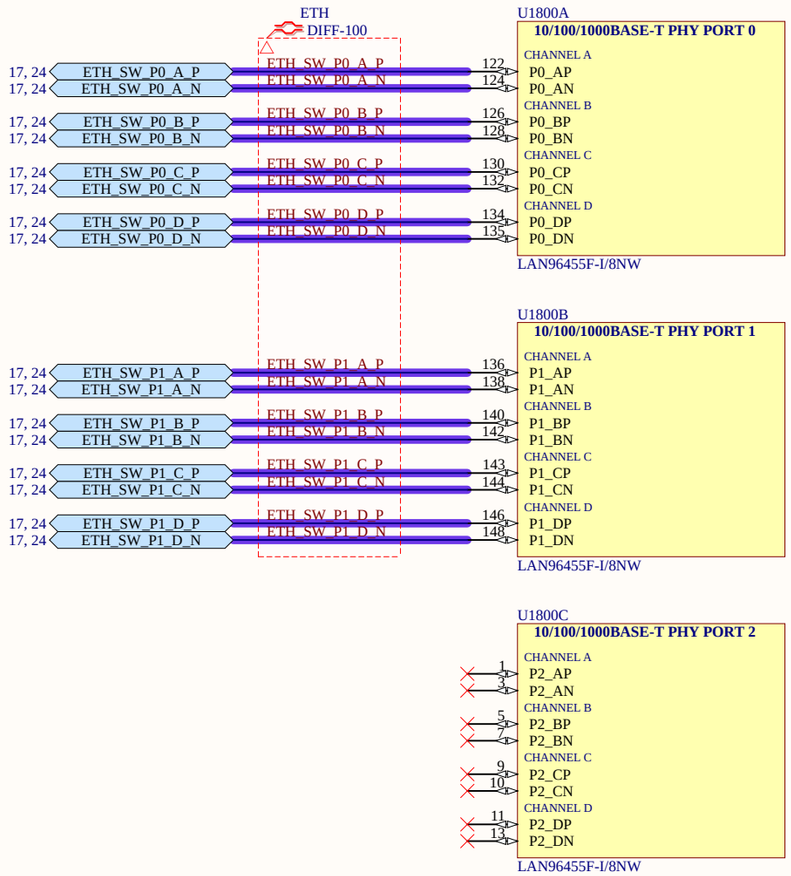


FIGURE 6-3: 1000MB COPPER PHY CONNECTIONS

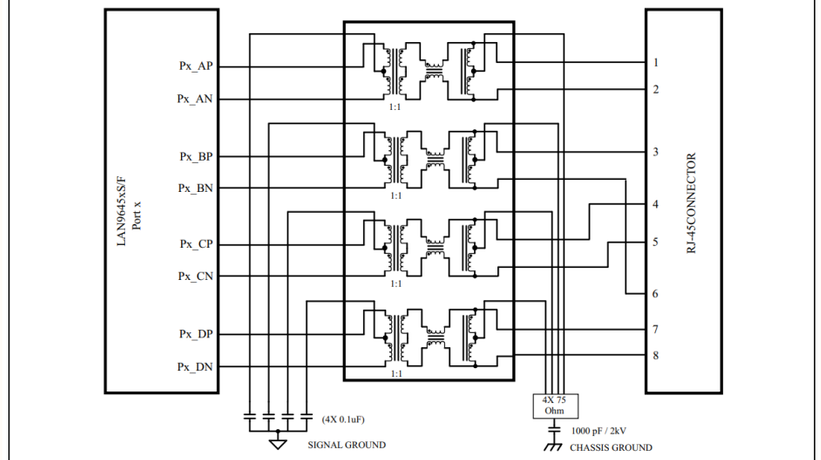
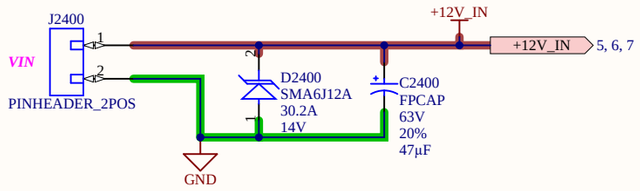
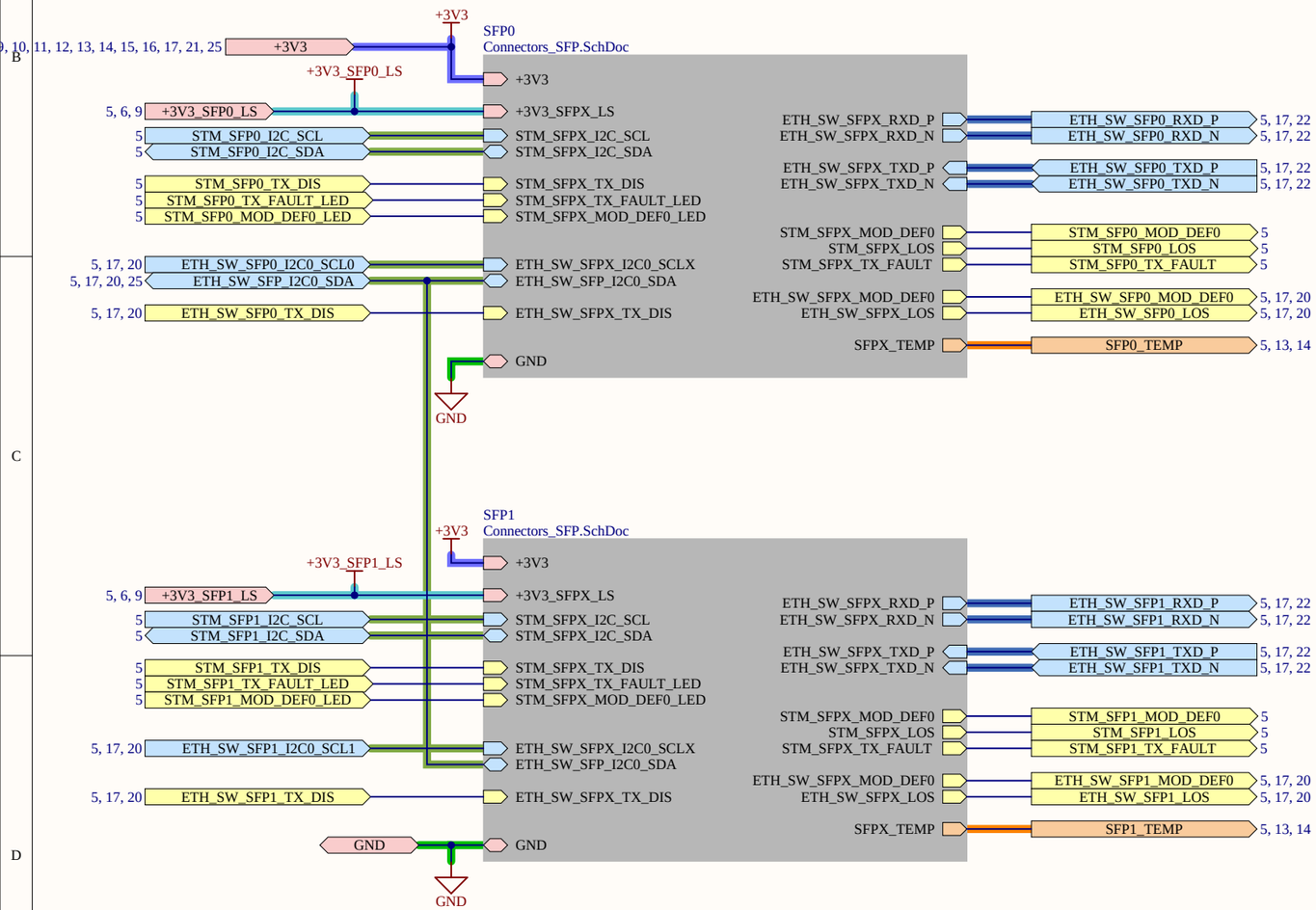


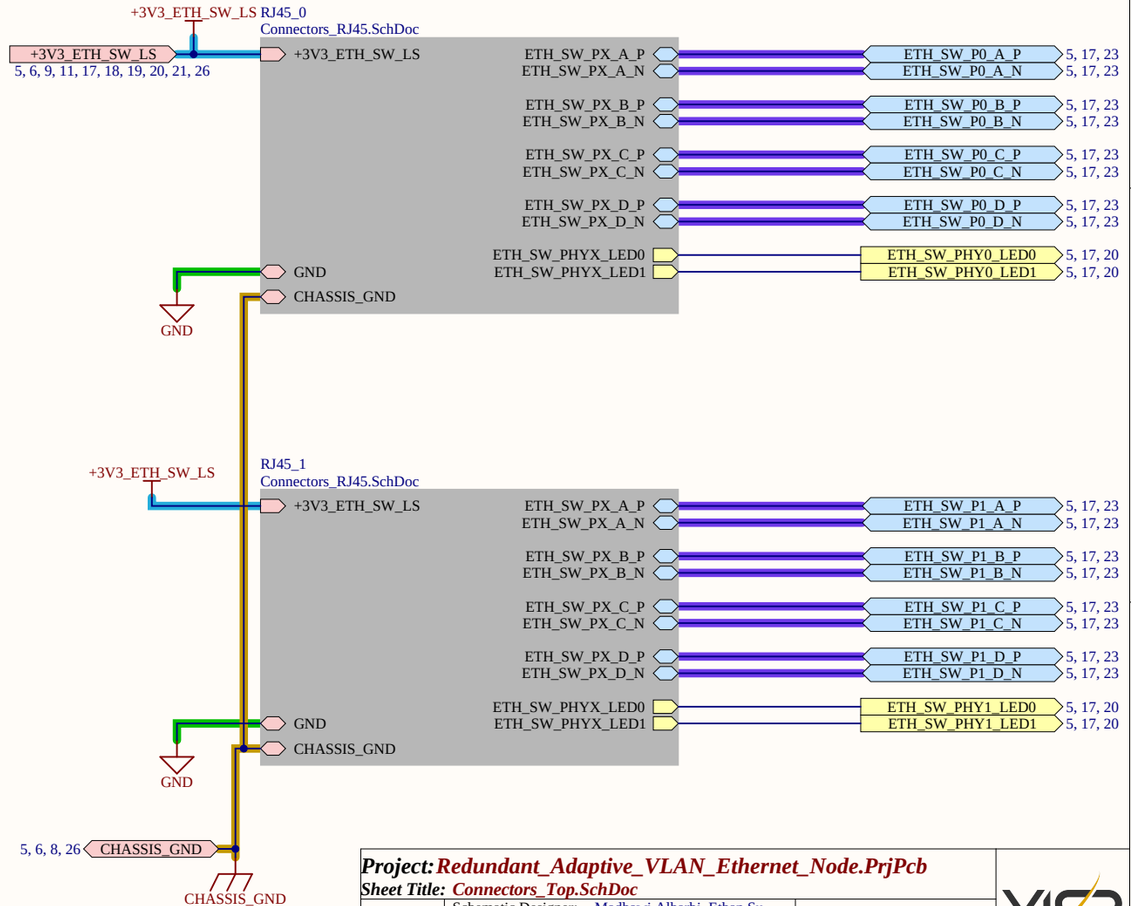
TABLE 4-4: LAN96455 PORT CONFIGURATIONS (5 PORTS)

LAN96455 Port Configuration	CuPHY					SGMII		RGMII	
	#0	#1	#2	#3	#4	#0	#1	#0	#1
5x CuPHY	DEV0	DEV1	DEV2	DEV3	DEV4	-	-	-	-
4x CuPHY + 1x 2.5G SGMII	DEV0	DEV1	DEV2	DEV3	-	DEV5	-	-	-
4x CuPHY + 1x RGMII	DEV0	DEV1	DEV2	DEV3	-	-	-	DEV7	-
3x CuPHY + 2x 2.5G SGMII	DEV0	DEV1	DEV2	-	-	DEV5	DEV6	-	-
3x CuPHY + 2x RGMII	DEV0	DEV1	DEV2	-	-	-	-	DEV7	DEV8
3x CuPHY + 1x 2.5G SGMII + 1x RGMII	DEV0	DEV1	DEV2	-	-	DEV5	-	DEV7	-
2x CuPHY + 1x 2.5G SGMII + 2x RGMII	DEV0	DEV1	-	-	-	DEV5	-	DEV7	DEV8
2x CuPHY + 2x 2.5G SGMII + 1x RGMII	DEV0	DEV1	-	-	-	DEV5	DEV6	DEV7	-
1x CuPHY + 2x 2.5G SGMII + 2x RGMII	DEV0	-	-	-	-	DEV5	DEV6	DEV7	DEV8

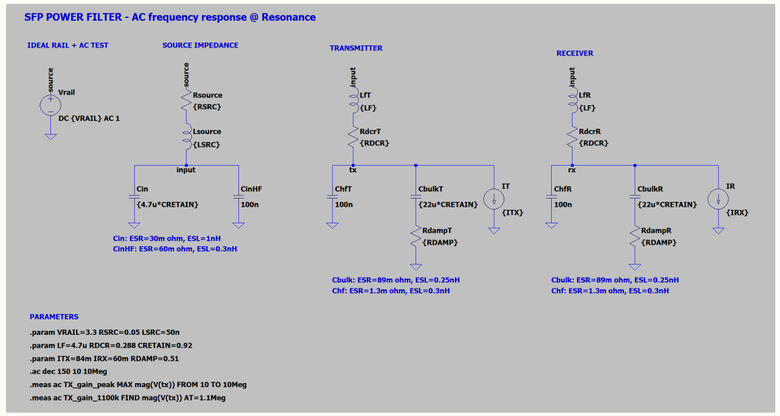
Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*		
Sheet Title: *ETH_SW_PORTS.SchDoc*		
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*
Date: 11/5/2026 Revision: 1.0	Sheet: 23 of: 26	



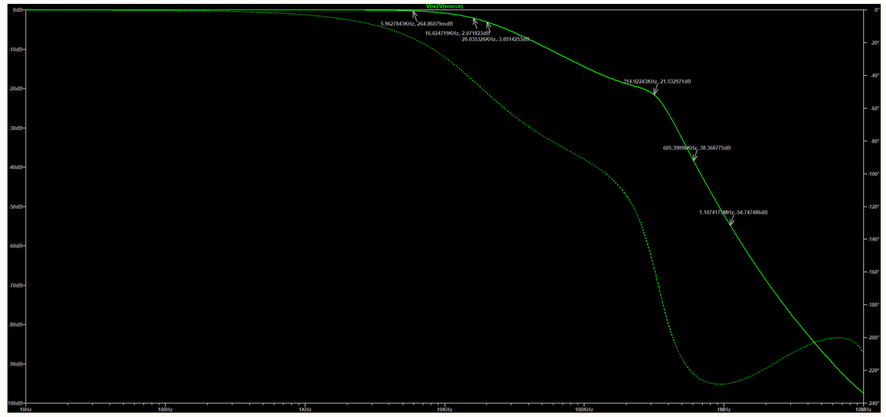




Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*			
Sheet Title: *Connectors_Top.SchDoc*			
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	
Date: 11/5/2026	Revision: 1.0	Sheet: 24 of: 26	



The frequency sweep covers low-frequency supply disturbances, the filter's natural-frequency region, the nominal 1.1 MHz switching frequency (3.3V buck) and its harmonics. The approximately 15.7 kHz nominal LC natural frequency is distinct from the buck switching frequency and capacitor self-resonance.



SFP Gigabit Ethernet Interface

1000BASE-X - Module Control & Status

DNP
R2500
0
0805
1%

Populate both filter branches as shown. Inductor or capacitor substitutions require review of DCR, current rating, effective capacitance at 3.3 V, ESR and filter response.

+3V3_SFPX_LS
24 +3V3_SFPX_LS
L2500
LQM21PN4R7MGRD
4.7µH
800mA
VCCT_LC
TP2500
VCCT_LC
C2500 0805 35V 20% 4.7µF
C2501 0603 100V 10% 100nF
C2504 0603 16V 20% 22µF
R2502 0.51 0805 1%
C2506 0603 100V 10% 100nF
GND

The 0.51 Ω resistors provide damping in series with the 22 µF capacitors. They are not series resistors in the DC supply path.

DNP
R2501
0
0805
1%
+3V3_SFPX_LS
VCCR_LC
TP2501
VCCR_LC
L2501
LQM21PN4R7MGRD
4.7µH
800mA
C2505 0603 16V 20% 22µF
C2507 0603 100V 10% 100nF
R2503 0.51 0805 1%
C2502 0603 100V 10% 100nF
C2503 0805 35V 20% 4.7µF
GND

Place each output 100 nF bypass capacitor close to its corresponding SFP supply pin, with a short ground return. Place the inductor and 22 µF/0.51 Ω damping branch nearby.

Avoid routing either VCCR_LC or VCCT_LC filtered supply near switching nodes.

J2500
VCCT
VCCR
VEET
VEER
0744410010

SFP DIFF-100
24 ETH_SW_SFPX_TXD_P — 0402 0 R2504 — SFPX_TXD_R_P 18 TD_P
24 ETH_SW_SFPX_TXD_N — 0402 0 R2506 — SFPX_TXD_R_N 19 TD_N
+3V3_SFPX_LS — R2508 10k 1% — SFPX_TX_DIS_R 3 TX_DISABLE — TP2502
GND — DNP R2510 10k 1% — 7 RATE_SELECT
+3V3 — R2512 10k 1% — TP2555 — SFPX_MOD_DEF0_R 6 MOD_DEF0
TP2506 — SFPX_I2C_SCL_R 5 MOD_DEF1_SCL
TP2507 — SFPX_I2C_SDA_R 4 MOD_DEF2_SDA
TX_FAULT 13
LOS 8
RD_P 12
RD_N 2
1 17 20 9 10 11 14
GND

SFP DIFF-100
SFPX_RXD_R_P — 0402 0 R2505 — ETH_SW_SFPX_RXD_P 24
SFPX_RXD_R_N — 0402 0 R2507 — ETH_SW_SFPX_RXD_N 24
STM_SFPX_TX_FAULT — R2509 10k 1% — +3V3_SFPX_LS
SFPX_LOS_R — R2511 10k 1% — +3V3_SFPX_LS
TP2503
STM_SFPX_TX_FAULT — TP2504 — STM_SFPX_TX_FAULT 24

Configure the connected LAN SerDes port for 1000BASE-X at 1.25 Gb/s serial rate for 1 Gb/s Ethernet operation.

Route TX± and RX± as 100 Ω differential pairs over a continuous ground reference. Preserve polarity and control intra-pair skew using the approved PCB impedance profile.

TX_FAULT is active HIGH. LOS is active HIGH and indicates loss of received optical signal. LOS LOW does not by itself confirm an established Ethernet link.

Handle insertion/removal and module startup before declaring stable status. Disable transmission when the module is absent. Use LAN port status for Ethernet link indication.

TEMP SNS

24, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 21, 24
+3V3 — +3V3 — U2500
VDD
VOUT — SFPX-TEMP — SFPX_TEMP 24
GND
50V 100nF — C2508 0402
TMP235A2DBZR

VOUT ≈ 0.500 V + 10 mV/°C × T
At T = 85°C:
VOUT = 0.5 + 0.01 × 85 = 1.35V
At T = 25°C --> VOUT = 0.75V

Place near source of heat. Measures local PCB temperature

Keep total Cload < 1000pF; TI suggests ~680 pF output filter as a starting point.

+3V3_SFPX_LS
R2513 4.7k 0603 1%
R2514 4.7k 0603 1%
SFPX_I2C_SDA_R
SFPX_I2C_SCL_R
24 STM_SFPX_I2C_SDA — 0603 0 R2515
24 STM_SFPX_I2C_SCL — 0603 0 R2516
24, 17, 20, 24 ETH_SW_SFP_I2C0_SDA — DNP 0603 0 R2517
24 ETH_SW_SFPX_I2C0_SCLX — DNP 0603 0 R2518

SFP0/1 share one SDA to ETH SW

J2501
SFPX_I2C_SCL — 3 SFPX_I2C_SCL_R
SFPX_I2C_SDA — 2 SFPX_I2C_SDA_R
GND — 1
SAMTEC_TSW-103-14-G-S
GND

Configure the STM TX_DISABLE GPIO as open-drain. Release the output to disable transmission; drive LOW to enable. Keep transmission disabled until module initialization is complete.

24 STM_SFPX_TX_DIS — 0603 0 R2521 — SFPX_TX_DIS_R
24 ETH_SW_SFPX_TX_DIS — DNP 0603 0 R2522

SFPX_LOS_R — 0603 0 R2519 — STM_SFPX_LOS 24
DNP 0603 0 R2520 — ETH_SW_SFPX_LOS 24
SFPX_MOD_DEF0_R — 0603 0 R2523 — STM_SFPX_MOD_DEF0 24
DNP 0603 0 R2524 — ETH_SW_SFPX_MOD_DEF0 24

+3V3 — R2525 1k 0603 1% — D2500 APT2012LZGCK — SFPX_MOD_DEF0_LED — TP2508 — STM_SFPX_MOD_DEF0_LED 24
SFPX_MOD_DFF0
+3V3 — R2526 1k 0603 1% — D2501 APT2012LSECK/J3-PRV — SFPX_TX_FAULT_LED — TP2509 — STM_SFPX_TX_FAULT_LED 24
SFPX_TX_FAULT

Initialize LED outputs OFF. Turn on the green LED when module presence is confirmed. Turn on the red LED when the module is present and TX_FAULT is asserted. Keep both LEDs OFF when the module is absent.

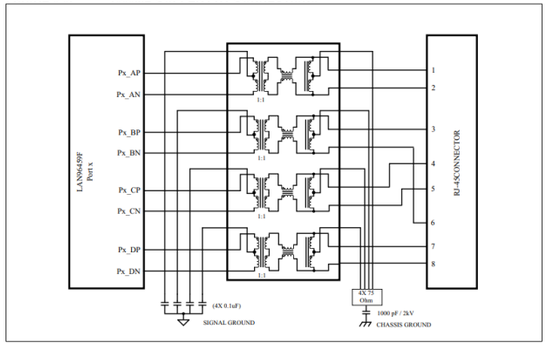
I. Pin Descriptions

Pin	Symbol	Name/Description	Ref.
1	V_{EET}	Transmitter Ground (Common with Receiver Ground)	1
2	T_{FAULT}	Transmitter Fault.	
3	T_{DIS}	Transmitter Disable. Laser output disabled on high or open.	2
4	MOD_DEF(2)	Module Definition 2. Data line for Serial ID.	3
5	MOD_DEF(1)	Module Definition 1. Clock line for Serial ID.	3
6	MOD_DEF(0)	Module Definition 0. Grounded within the module.	3
7	Rate Select	No connection required	
8	LOS	Loss of Signal indication. Logic 0 indicates normal operation.	4
9	V_{EER}	Receiver Ground (Common with Transmitter Ground)	1
10	V_{EER}	Receiver Ground (Common with Transmitter Ground)	1
11	V_{EER}	Receiver Ground (Common with Transmitter Ground)	1
12	RD-	Receiver Inverted DATA out. AC Coupled	
13	RD+	Receiver Non-inverted DATA out. AC Coupled	
14	V_{EER}	Receiver Ground (Common with Transmitter Ground)	1
15	V_{CCR}	Receiver Power Supply	
16	V_{CCT}	Transmitter Power Supply	
17	V_{EET}	Transmitter Ground (Common with Receiver Ground)	1
18	TD+	Transmitter Non-Inverted DATA in. AC Coupled.	
19	TD-	Transmitter Inverted DATA in. AC Coupled.	
20	V_{EET}	Transmitter Ground (Common with Receiver Ground)	1

Project: *Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb*
Sheet Title: *Connectors_SFP.SchDoc*

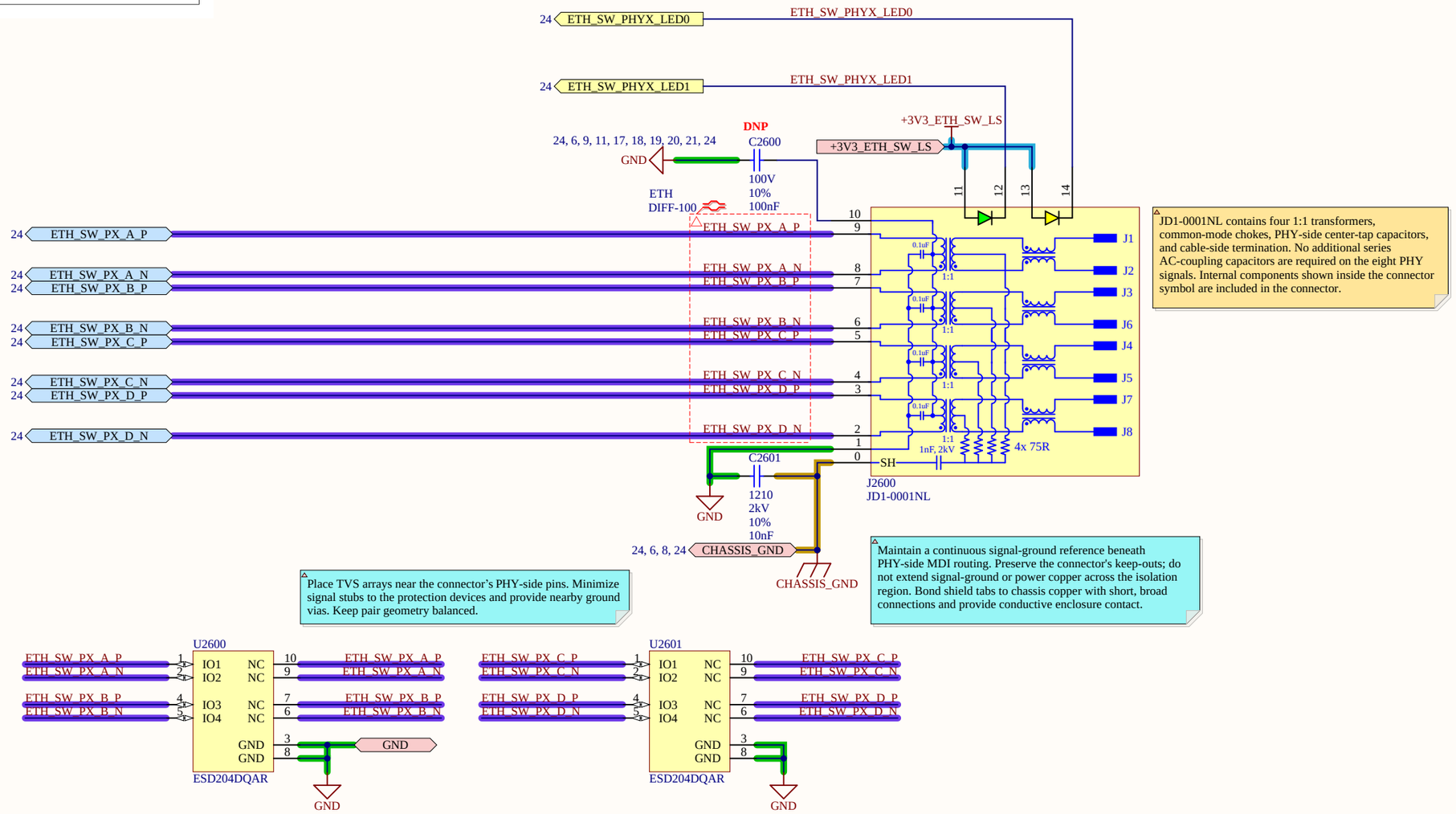
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*
Date: 11/5/2026 Revision: 1.0	Sheet: 25 of: 26	

FIGURE 6-3: 1000MB COPPER PHY CONNECTIONS



RJ45 Gigabit Ethernet Interface

10/100/1000BASE-T w/ Integrated Magnetics



Project:	***Redundant_Adaptive_VLAN_Ethernet_Node.PrjPcb***		
Sheet Title:	***Connectors_RJ45.SchDoc***		
Size: A4	Schematic Designer: Madhawi Alharbi, Ethan Su PCB Layout Designer: Madhawi Alharbi Responsible Engineer: Madhawi Alharbi	*Yellow Jacket Space Program* *Georgia Tech* *Atlanta, GA*	
Date: 11/5/2026	Revision: 1.0	Sheet: 26 of: 26	